Neuroscience

Dissociable memory modulation mechanisms facilitate fear amnesia at different timescales

Reviewed Preprint
v2 • March 12, 2025
Revised by authors

Reviewed Preprint
v1 • July 8, 2024

Yinmei Ni, Ye Wang, Zijian Zhu, Jingchu Hu, Daniela Schiller, Jian Li

School of Psychological and Cognitive Sciences and Beijing Key Laboratory of Behavior and Mental Health, Peking University, Beijing, China • School of Psychology, Shaanxi Normal University, Xi'an, China • Department of Anxiety Disorders, Shenzhen Kangning Hospital, Shenzhen Mental Health Institute, Shenzhen, China • Departments of Psychiatry and Neuroscience, and Friedman Brain Institute, Icahn School of Medicine at Mount Sinai, New York, United States • IDG/McGovern Institute for Brain Research, Peking University, Beijing, China

eLife Assessment

The **valuable** findings in this study reveal an intricate pattern of memory expression following retrieval extinction at different intervals from retrieval-extinction to test. They document that immediately after extinction there is a nonselective impairment in memory, which leads to no impairment at a 6-hour interval. At a 24-hour interval, there is a selective impairment. The evidence supporting the claims is **incomplete** and there are inconsistencies in the analyses reported that obscure the interpretation.

https://doi.org/10.7554/eLife.98652.2.sa3

Abstract

Memory reactivation renders consolidated memory fragile and sets the stage for memory reconsolidation. However, whether memory retrieval facilitates update mechanisms other than memory reconsolidation remains unclear. We tested this hypothesis in three experiments with healthy human participants. First, we demonstrate that memory retrieval-extinction protocol prevents the return of fear expression shortly after extinction training and this short-term effect is memory reactivation dependent (Study 1, N = 57 adults). Furthermore, across different timescales, the memory retrieval-extinction paradigm triggers distinct types of fear amnesia in terms of cue-specificity and cognitive control dependence, suggesting that the short-term fear amnesia might be caused by different mechanisms from the cue-specific amnesia at a longer and separable timescale (Study 2, N = 79 adults). Finally, using continuous theta-burst stimulation (Study 3, N = 75 adults), we directly manipulated brain activity in the dorsolateral prefrontal cortex, and found that both memory reactivation and intact prefrontal cortex function were necessary for the short-term fear amnesia after the retrieval-extinction protocol. The differences in temporal scale, cue-specificity, and cognitive control ability dependence between the short- and long-term amnesia suggest that memory retrieval and extinction training trigger distinct underlying memory update mechanisms. These findings suggest the potential involvement of coordinated memory

modulation processes upon memory retrieval and may inform clinical approaches for addressing persistent maladaptive memories.

Introduction

Memory retrieval provides a unique window through which consolidated memories can be modified or even neutralized. Studies of memory reconsolidation suggest that memories are rendered fragile each time they are retrieved (Alberini, 2005; Dudai, 2006; Nader, Schafe, & LeDoux, 2000), possibly due to the disruption of the old memory trace. The reconsolidation of the new memory trace also requires *de novo* protein synthesis, which usually takes hours to complete. Pharmacological blockade of protein synthesis and behavioral interventions can both eliminate the original fear memory expression in the long-term (24 hours later) memory test (Lee, 2008; Lee et al., 2017; Schiller et al., 2013; Schiller et al., 2010), resulting in the cue-specific fear memory deficit (Debiec et al., 2002; Lee, 2008; Nader, Schafe, & LeDoux, 2000). For example, during the reconsolidation window, retrieving a fear memory allows it to be updated through extinction training (i.e., the retrieval-extinction paradigm (Agren, 2014; Agren et al., 2012; Lee, 2008; Lee et al., 2017; Schiller et al., 2013; Schiller et al., 2010), but also see (Chalkia, Schroyens, et al., 2020; Chalkia, Van Oudenhove, & Beckers, 2020; Schiller, 2020).

However, the exact functional role of memory retrieval remains unclear. For example, while previous fear memory studies typically reported the reconsolidation effects emerging several hours later (the long-term effect) (Kindt and Soeter, 2018; Lee, 2008; Speer et al., 2021), the temporal dynamics of the amnesia effect through behavioral manipulation such as retrieval-extinction is often overlooked in the fear memory literature (Lee et al., 2017; Schiller et al., 2013; Schiller et al., 2010). It is possible that memory retrieval might facilitate memory modification at different temporal scales. This is in stark contrast to semantic and episodic memory studies where the behavioral manipulations such as the retrieval-induced forgetting (RIF) protocol or direct suppression result in short-term memory deficit within 30 minutes (Anderson et al., 2000; Anderson et al., 1994). More recently, behavioral strategies originally developed in the declarative memory studies such as direct suppression had also been successfully applied to the fear memory paradigms in humans and prevented the return of fear shortly afterward (30mins), suggesting that the fear memory might be amenable to modulation and this effect can manifest at a more immediate timescale, to which the memory reconsolidation theory remain agnostic (Wang et al., 2021).

Corresponding to the long-term behavioral manifestation, concurrent neural evidence supporting memory reconsolidation hypothesis emphasizes that synapse degradation and *de novo* protein synthesis are required for reconsolidation. These processes typically take several hours, if not longer, and are often aided by overnight sleep (Kindt and Soeter, 2018). On the contrary, previous behavioral manipulations engendering the short-term amnesia on declarative memory, such as the think/no-think paradigm, hinges on the intact activities in brain areas such as dorsolateral prefrontal cortex (cognitive control) and its functional coupling with specific brain regions such as hippocampus (memory retrieval) (Anderson and Green, 2001; Wimber et al., 2015). The declarative amnesia effect arises much earlier due to the more instant modulation of functional connectivity, rather than the slower processes of new protein synthesis in these brain regions (Wang et al., 2021). However, it remains unclear whether experimental paradigms such as memory retrieval-extinction, a method developed to induce long-term prevention of fear expression based on the memory reconsolidation hypothesis, might also trigger short-term amnesia for the fear memory.

To fully comprehend the temporal dynamics of the memory retrieval-extinction training effect, we first carried out a simple fear memory study by pairing a neutral cue (conditioned stimulus, CS) with the electric shock (unconditioned stimulus, US) to directly test whether the fear retrieval-extinction paradigm would yield a short-term effect on fear expression. We then performed another double cue CS-US fear memory experiment to study the temporal characteristics of fear memory malleability to behavioral extinction at the short- (30mins), medium- (6 hours) and long-term (24 hours) time scales.

We hypothesize that the labile state triggered by the memory retrieval may facilitate different memory update mechanisms following extinction training, and these mechanisms can be further disentangled through the lens of temporal dynamics and cue-specificities. Specifically, memory reconsolidation effect, if exists, should only be evident in the long-term (24h) memory test due to the requirement of new protein synthesis and should be cue-dependent (Monfils et al., 2009). In contrast, if more immediate memory update mechanisms are involved, then the fear memory deficit should be observed relatively early (30mins) after extinction training (Anderson and Floresco, 2022; Anderson and Green, 2001). There may even exist a "limbo" state: when the immediate memory update effects dissipate and the reconsolidation updating is yet to take effect (6h), the conditioned stimuli should still elicit fear responses, regardless of the reminder (cue). Previous research in the short-term declarative and fear memory deficits highlighted the role of cognitive control. Subjects with stronger control over intrusive thoughts or stronger functional connectivity between brain regions such as the dorsolateral prefrontal cortex (dlPFC) and hippocampus exhibited more pronounced amnesia in the memory suppression tasks (Anderson et al., 1994; Wells and Davies, 1994; Wimber et al., 2015). We reason that if the retrieval related short-term fear amnesia is also related to cognitive control and the dlPFC activities, then the individual difference of such immediate fear memory deficits should be associated with individual subject's control abilities over intrusive thoughts (Küpper et al., 2014), which can be measured via the Thought-Control Ability Questionnaire (TCAQ) scale.

Indeed, through a series of experiments, we identified a short-term fear amnesia effect following memory retrieval-extinction training, in addition to the long-term amnesia effect that appeared much later. Finally, by applying the continuous theta-burst stimulation (cTBS) protocol over the dlPFC, the brain region critical for cognitive control and goal maintenance, we showed that both memory retrieval and intact dlPFC activity were necessary for the more immediate fear amnesia effect.

Open Practices Statement

Neither of the studies reported in this article was preregistered. The data for both studies are publicly accessible at *https://osf.io/9agvk/*.

Methods

Participants

All participants were students recruited from Peking University. They were right-handed with normal vision and had not participated in electric shock-related experiments before. None of the participants had current or history of psychiatric illness, nor any contraindication on the application of transcranial magnetic stimulation (TMS) (Rossi et al., 2009; Rossini et al., 2015). All participants provided informed consent and were remunerated for their participation. This study was approved by the institutional review board of school of psychological and cognitive sciences at Peking University.

We first conducted a power analysis (G*Power) to determine the number of participants sufficient to detect a reliable effect (Faul et al., 2009). Based on the reported effect size of the reinstated fear memory between the treatment and control groups on fear memories reported in the previous literature (η^2 = 0.19, 0.216, 0.24; N = 40, 55, 30 of sample size, respectively (Kindt and Soeter, 2018; Kindt et al., 2009; Wang et al., 2021)), 52 participants for both groups (study 1:

reminder group and no-reminder group) were enough to detect a significant interaction effect (α = 0.05, β = 0.9, 2 (groups) × 2 (phases) two-way ANOVA interaction effect). Similarly, a total of 72 participants for three groups (study 2: 30-min group, 6-h group and 24-h group) were required to detect a reliable effect (α = 0.05, β = 0.9, 3 (groups) × 2 (phases) × 2 (CS+) three-way ANOVA interaction effect) (Heo and Leon, 2010). For the rTMS study, based on the reported effect size of reinstated fear memory (η^2 = 0.144, 0.19, 0.216, 0.24; N =79, 40, 55, 30 total sample size, respectively (Kindt and Soeter, 2018; Kindt et al., 2009; Wang et al., 2021)), a total of 72 participants (study 3) were needed to detect a significant interaction effect (α = 0.05, β = 0.8, 4 (groups) × 2 (phases) × 2 (CS+) three-way ANOVA interaction effect).

In study 1, we first recruited a total of 75 human subjects (41 females; mean age= 21.3, SD= 2.21). Eight participants discontinued after Day 1 due to their non-measurable spontaneous conditioned stimulus (CS) signal (the skin conductance response, SCR) towards different CSs (non-responders, SCR response to any CS was less than 0.02 μS, n = 0 & 8; reminder group and no-reminder group, respectively). Ten participants (7 females; mean age = 20.6, SD = 1.79) who were "non-learners" during fear acquisition (n = 3 & 1 in both groups; respectively) or extinction (n = 3 & 3, respectively) phase were also excluded from further analysis. The criteria for "non-learners" in the fear acquisition include smaller mean SCR of CS+ (CS that was partially paired with the electric shock) than that of the CS− (CS that was never paired with shock) in the trials in the latter half of acquisition, and a smaller mean SCR difference between CS+ and CS− in the latter than the first half trials of fear acquisition on day 1. Similarly, the criteria for "non-learners" in the fear extinction are larger mean SCRs of CS+ than the CS− responses in both the last trial and the latter-half trials of extinction and a larger mean SCR difference between CS+ and CS− in the latter than the first half trials during fear extinction on day 2 (see the exclusion criteria Table S1 in the Supplemental Material). Therefore, we had a total of 57 subjects for data analysis in study 1 (30 for the reminder group, 17 females, mean age = 21.4, SD = 2.37, and 27 for the no-reminder group, 15 females, mean age = 21.62, SD = 2.13). The group × gender interaction effect on participants' age was not statistically significant ($F_{1,49}$ = 0.137, P = 0.713, η^2 = 0.003, two-way ANOVA), and gender (χ^2_1 = 0.007, P = 0.933) and age (t_{55} = −0.376, P = 0.708) were not significantly different between two groups.

In study 2, we first recruited 119 participants (56 females; mean age= 20.85, SD= 2.59). Thirty-seven participants discontinued after Day 1 for they were "non-responders" (n = 19, 5 and 13; 30-min group, 6-h group and 24-h group, respectively). Additionally, two participants (n = 1&1; 30-min group and 24-h group; respectively) failed to show the evidence of fear acquisition and one participant (30-min group) failed to show the evidence of fear extinction on day 2 and were also excluded from further analysis. Finally, we had a total of 79 subjects for data analysis in study 2 (27 for the 30-min group, 15 females, mean age = 21.4, SD = 2.46; 26 for the 6-h group, 12 females, mean age = 20.54, SD = 1.74; and 26 for the 24-h group, 12 females, mean age = 20.96, SD = 2.990). The interaction effect of gender × group on participants' age was not statistically significant ($F_{2,73}$ = 0.213, P = 0.809, η^2 = 0.006, two-way ANOVA), and gender (χ^2_2 = 0.021, P = 0.990) or age ($F_{2,76}$ = 0.661, P = 0.519, η^2 = 0.017) was not significantly different across three groups.

In study 3, we recruited a total of 90 participants (47 females; mean age = 21.01, SD = 2.44). Fifteen participants (8 females; mean age = 21.2, SD = 1.22) were excluded from further analysis since they failed to show SCR response to any stimulus (CS1+, CS2+ or CS−) (non-responders, n = 3,4,4,4; R rdlPFC group, R vertex group, NR rdlPFC group and NR vertex group, respectively, see Supplemental Table S1) (de Voogd and Phelps, 2020; Dunsmoor et al., 2015; Dunsmoor et al., 2017; Hu et al., 2018; Raio et al., 2017; Schiller et al., 2013; Schiller et al., 2010; Schiller et al., 2012). Our final sample included 75 participants for data analysis (19 for the Reminder right dlPFC group, 10 females, mean age = 22.05, SD = 2.44; 18 for the Reminder vertex group, 9 females, mean age = 20.67, SD = 2.68; 18 for the No-reminder right dlPFC group, 10 females, mean age = 20.28, SD = 2.85; and 20 for the No-reminder vertex group, 10 females, mean age = 20.85, SD = 2.43).

Conditioned stimuli and psychophysiological stimulation

For study 1, we employed two squares with different colors (yellow and blue) that served as CS+ and CS−. During fear acquisition, the CS+ was paired with a mild electrical shock (US) on a 37.5% partial reinforcement scheme (6 CS+ paired with electric shock and 10 CS+ with no shock), and the CS− was never paired with a shock (10 CS−). A pseudo-random CS delivery order was generated for the fear acquisition, extinction and test phase for two groups with the rule that no same trial-type (CS+ and CS−) repeated more than twice. Assignment of square colors to the CS (CS+ and CS−) was counterbalanced across participants.

For study 2 & 3, three squares with different colors (yellow, red and blue) were used as CS1+, CS2+ and CS−, respectively. During the fear acquisition phase, both CS1+ and CS2+ were paired with electric shocks (US) on a 37.5% partial reinforcement scheme, and the CS− was not paired with shocks (10 non-reinforced presentations of CS1+, CS2+ and CS− each, intermixed with an extra of 6 CS1+ and 6 CS2+ that co-terminated with the shock). Again, a pseudo-random stimulus order was generated for fear acquisition and extinction phases of three groups with the rule that no same trial-type (CS1+, CS2+ and CS−) repeated more than twice. In the test phase, to exclude the possibility that the difference between CS1+ and CS2+ was simply caused by the presentation sequence of CS1+ and CS2+, half of the participants completed the test phase using a pseudo-random stimuli sequence and the identities of CS1+ and CS2+ reversed in the other half of the participants. For all three groups, the CS colors were counterbalanced across participants.

The US was a mild electrical shock delivered to the right wrist of participants via a DS-5 Isolated Bipolar Constant Current Stimulator (Digitimer, Welwyn Garden City, UK). The US levels were set by the participants themselves, starting with a low shock level (5v) and gradually increased and settled to a level that they described as 'uncomfortable, but not painful' (with the maximum level of 10v). The duration of all electric shocks was 200ms with 50 pulses per second.

SCR measurements were collected using two Ag-AgCI electrodes attached to the tips of the index and middle fingers of each subject's left hand. All skin conductance data were recorded via the Biopac® MP160 BioNomadix System and analyzed using the Acknowledgement 5.0 software. The SCR level for each trial was calculated as the amplitude difference (in micro siemens) from peak to trough during the 0.5 to 4.5s window after the CS (CS SCR) or US (US SCR) onset and responses below 0.02μS were encoded as zero. The raw SCR scores were divided by each subject's mean US responses and then square-root-transformed to normalize SCR across individuals and groups (Wang et al., 2021).

Continuous theta burst stimulation (cTBS), a specific form of repetitive transcranial magnetic stimulation (rTMS), was applied with a Magstim Rapid 2 system (Magstim, Whitland, Wales, UK). Before the experiments, we determined the stimulation intensity of the cTBS protocol that was at 80% of the individual active motor threshold (AMT). Each transcranial magnetic stimulation (TBS) burst consisted of three stimuli pulses at 50HZ, with each train being repeated every 200ms (5Hz). In this cTBS protocol, a 40s train of uninterrupted TBS is given (600 pluses) (Borgomaneri et al., 2020a; Su et al., 2022). During the stimulation, the cTBS was applied either over the right dlPFC or the vertex, which was determined by standard F4 and Cz locations based on international electroencephalogram 10-20 system measurements (Censor et al., 2010).

The control ability over intrusive thoughts was measured by the 25-item Thought-Control Ability Questionnaire (TCAQ) scale (Wells and Davies, 1994). Participants were asked to rate on a five-point Likert-type scale the extent to which they agreed with the statement from 1 (*completely disagree*) to 5 (*completely agree*). At the end of the experiments, all participants completed the TCAQ scale to assess their perceived control abilities over intrusive thoughts in daily life (Küpper et al., 2014). The internal consistency (Cronbach's alpha) of TCAQ scale was 0.92, and the reliability coefficient was satisfactory (r = 0.88).

To measure the extent to which fear returns after the presentation of unconditioned stimuli (US, electric shock) in the test phase, we defined the fear recovery index as the SCR difference between the first test trial and the last extinction trial for a specific CS for each subject. Similarly, in studies 2 and 3, differential fear recovery index was defined as the difference between fear recovery indices of CS+ and CS− for both CS1+ and CS2+.

Experimental procedure

Study 1, 2 and 3 were carried out for two or three consecutive days with three phases. Before the experiments, all participants gave informed consent. During the experiments, subjects were required to stay relaxed and still, focus on the computer screen and pay attention to the relationship between color stimuli (CS) and the electric shock (US).

Day 1: Acquisition. For study 1, two groups of participants underwent a fear-conditioning paradigm (acquisition) where the CS+ was partially (37.5%) paired with the US and the CS− was never paired with the shock. For study 2 & 3, participants underwent fear conditioning using three color squares as the CSs (CS1+, CS2+ and CS−). The CSs were presented for 4s each with a 10-12s variable ITI in all the studies (1, 2 and 3).

Day 2: Retrieval-extinction phase. For study 1, the fear memory was reactivated before extinction in the reminder group. During reactivation, the CS+ was presented once without the delivery of the US. Followed by a 10-min break, the reminder group underwent extinction training with 10 CS+ and 11 CS− presentation non-reinforced. In the no-reminder group, the fear memory was not reactivated (no CS+ reminder). To match with the timeline of the reminder group, the no-reminder group subjects still needed to wait for 10-min after the beginning of the experiment on day 2 and then underwent extinction training with 11 CS+ and 11 CS+ non-reinforced presentation. During the 10-min break, all participants were asked to rest and stay still.

For study 2, only one of the CS+ (CS1+) was presented once without the US during reactivation in all three groups. After a 10-min break, all three groups underwent extinction training with 10 CS1+, 11CS2+ and 11CS− non-reinforced presentation.

For study 3, for the reminder groups (reminder-dlPFC and reminder-vertex groups), only one of the CS+ (CS1+) was presented once (without the US delivery) as the retrieval trial. No CS+ reminder was delivered for the no-reminder groups (no-reminder dlPFC and no-reminder vertex groups). Additionally, during the 10-min break (8 mins after the supposed CS+ reminder delivery), cTBS was applied either over the right dlPFC (dlPFC groups: PFC) or the vertex (vertex groups: VER), resulting in a 2 (reminder vs. no-reminder) x 2 (dlPFC vs. vertex) of 4 different groups (reminder-dlPFC: R-PFC, reminder-vertex: R-VER, no-reminder dlPFC: NR-PFC, no-reminder vertex: NR-VER).

Day 2 or Day 3: Reinstatement and test phases. For study 1, 30 minutes after the extinction training, both groups received four un-signaled electric shocks with 10s to 12s ITI (reinstatement). Eighteen seconds after the reinstatement phase, all subjects were presented with CS+ and CS− 10 times each without electric shocks and their SCRs were recorded (test phase).

For study 2, participants were assigned into three groups: the 30min group, the 6h group and the 24h group. All participants received four un-signaled electric shocks with 10s to 12s ITI (reinstatement) after their corresponding time delays (30min, 6h or 24h) between extinction and reinstatement. Eighteen seconds after the reinstatement phase, all subjects were tested via the presentation of CS1+, CS2+ and CS− 10 times each without electric shocks associated.

For study 3, the reinstatement and testing procedures were the same as in study 2 except that they were conducted one hour after the extinction training.

Results

The short-term effect of the fear memory retrieval-extinction procedure

To test the memory-retrieval-induced fear memory deficits at different time scales in humans, we designed two experiments to examine whether the fear memory retrieval-extinction procedure would block the return of fear memory.

In the first study, we aimed to test whether there is a short-term amnesia effect following the fear retrieval-extinction paradigm. We measured subjects' SCRs for the fear acquisition, extinction and test phases to assess the associative fear memory and the recovery of fear memory. At each phase, the differential fear response was calculated as the difference between SCRs to the CS+ and CS−.

To assess fear acquisition across groups (**Figure 1B** and **C**), we conducted a mixed two-way ANOVA of group (reminder vs. no-reminder) x time (early vs. late part of the acquisition) on the differential fear SCR. Our results showed a significant main effect of time (early vs. late; $F_{1,55}$ = 6.545, P = 0.013, η^2 = 0.106), suggesting successful fear acquisition in both groups. There was no main effect of group (reminder vs. no-reminder) or the group x time interaction (group: $F_{1,55}$ = 0.057, P = 0.813, η^2 = 0.001; interaction: $F_{1,55}$ = 0.066, P = 0.798, η^2 = 0.001), indicating similar levels of fear acquisition between two groups. Post-hoc t-tests confirmed that the fear responses to the CS+ were significantly higher than that of CS− during the late part of acquisition phase in both groups (reminder group: t_{29} = 6.642, P < 0.001; no-reminder group: t_{26} = 8.522, P < 0.001; **Figure 1C**). Importantly, the levels of acquisition were equivalent in both groups (early acquisition: t_{55} = −0.063, P = 0.950; late acquisition: t_{55} = −0.318, P = 0.751; **Figure 1C**).

To ensure equivalent and successful extinction across groups (**Figure 1B** and **D**), we performed similar mixed two-way ANOVA (group x trial) on the differential SCR in the extinction phase. Our results showed significant main effect of trial ($F_{10,550}$ = 2.825, P = 0.010, η^2= 0.049), but no effect of group (F = 0.063, P = 0.802, η^2 = 0.001) or their interaction (F = 1.77, P = 0.101, η^2 = 0.031). Follow-up t-tests further confirmed that in the last trial of extinction there was no difference between fear responses to CS+ and CS− within group (reminder group: t_{29} = −1.147, P = 0.261; no-reminder group: t_{26} = 0.261, P = 0.796; **Figure 1D**). Importantly, the levels of extinction did not show significant difference between two groups (t_{55} = −1.074, P = 0.288; **Figure 1D**).

Fear recovery was assessed using a two-way ANOVA with main effects of group (reminder vs. no-reminder) and time (last trial of extinction vs. first trial of test) and it showed a significant time × group interaction ($F_{1,55}$ = 4.087, P = 0.048, η^2= 0.069). We defined the fear recovery index as the SCR difference between the first test trial and the last extinction trial for a specific CS. The interaction effect was confirmed by the significant difference between the differential fear recovery indexes between CS1+ and CS2+ in the reminder and no-reminder groups (t_{55} = −2.022, P = 0.048; **Figure 1E**, also see Supplemental Material for the direct test of the test phase). Post hoc t-tests showed that fear memories were resilient after regular extinction training, as demonstrated by the significant difference between fear recovery indexes of the CS+ and CS− for the no-reminder group (t_{26} = 7.441, P < 0.001; Bonferroni correction, **Figure 1E**), while subjects in the reminder group showed no difference of fear recovery between CS+ and CS− (t_{29} = 0.797, P = 0.432, **Figure 1E**). Also, our results remain robust even with the "non-learners" included in the analysis (Figure supplement 1 in the Supplemental Material).

Together, these results indicate that the retrieval-extinction procedure prevents the return of fear expression in the short-term test, a phenomenon distinct from the hypothesized fear reconsolidation effect at a much later time (Schiller et al., 2013). In the current study, fear

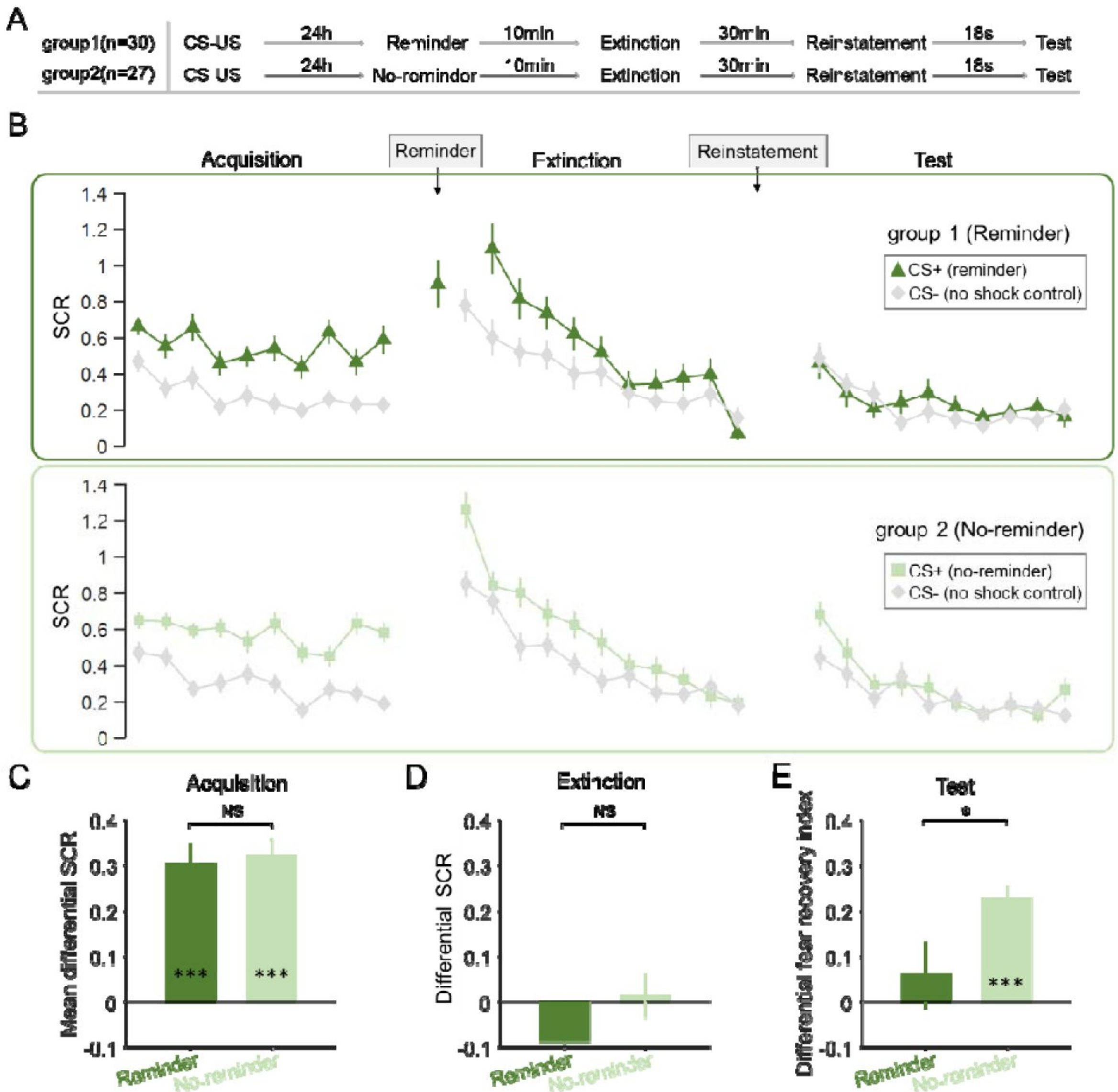


Figure 1|

SCR responses to conditioned stimuli in two groups of study 1 indicating short-term memory amnesia (**A**) Experimental design and timeline. (**B**) SCRs of fear conditioned stimuli (CS+) and the control stimulus (CS−) across fear acquisition, extinction and test phases for two groups (reminder and no-reminder). (**C**) Mean differential SCRs (CS+ minus CS−) in the acquisition phase (trials in the latter half). (**D**) Differential SCRs (CS+ minus CS−) in the extinction phase (last trial). **(E)** Differential fear recovery index between CS+ and CS− in the test phase. ****P* < 0.001, **P* < 0.05. NS: Non-significant. Error bars represent standard errors.

recovery was tested 30 minutes after extinction training, whereas the effect of memory reconsolidation was generally evident only several hours later and possibly with the help of sleep (Kindt and Soeter, 2018; Speer et al., 2021), leaving open the possibility of a different cognitive mechanism for the short-term fear amnesia related to the retrieval-extinction procedure. Importantly, such a short-term effect is also retrieval dependent, suggesting the labile state of memory may be necessary for the short-term memory update to take effect (**Figure 1E**).

Temporal and generalizing characteristics of the retrieval-extinction effect

Given the findings of short-term fear amnesia following the retrieval-extinction paradigm, we set out to map the temporal dynamics of fear amnesia as well as its cue-specificities - whether the amnesia triggered by a specific CS+ reminder (for example CS1+) generalizes to the fear memory associated with the other CS+ (CS2+). We adopted a double-cue fear learning paradigm and examined fear recovery in the short- (30 mins), medium- (6 hours) and long-term (24 hours) fear memory tests in three separate groups of participants after the extinction training (**Figure 2A**).

Fear acquisition was assessed using a mixed three-way ANOVA with group (30min, 6h and 24h), CS+ (CS1+ vs. CS2+, defined as the mean SCR differences between each CS+ and CS−) and trial numbers (10 trials) as main factors. This analysis showed a significant main effect of trial ($F_{9,684}$ = 12.782, P < 0.001, η^2 = 0.144), but no effect of group ($F_{2,76}$ = 2.121, P = 0.127, η^2 = 0.053), CS+ ($F_{1,76}$ = 0.576, P = 0.45, η^2 = 0.008) or their interactions (all Ps > 0.05; **Figure 2B**). Furthermore, to ensure equivalent fear acquisition across three groups, a mixed two-way ANOVA showed no effect of group (30min, 6h and 24h; $F_{2,76}$ = 1.79, P = 0.174, η^2 = 0.045), CS+ (CS1+ vs. CS2+; $F_{1,76}$ = 1.337, P = 0.251, η^2 = 0.017) or their interaction ($F_{2,76}$ = 0.959, P = 0.388, η^2 = 0.025) on the late phase of acquisition (last 5 trials). Post-hoc t-tests confirmed that the SCRs elicited by CS1+ and CS2+ were significantly higher than that of CS− among three groups on the late phase of acquisition (CS1+: t_{26} = 6.806, P < 0.001; CS2+: t_{26} = 6.854, P < 0.001 for the 30min group; CS1+: t_{25} = 9.822, P < 0.001; CS2+: t_{25} = 9.841, P < 0.001 for the 6h group and CS1+: t_{25} = 6.909, P < 0.001; CS2+: t_{25} = 6.956, P < 0.001 for the 24h group), indicating successful and similar fear acquisition across all groups (**Figure 2C**).

We then set out to examine whether participants underwent successful extinction training on day 2. The similar mixed three-way ANOVA showed a significant effect of trial ($F_{10,760}$ = 8.220, P < 0.001, η^2 = 0.098), but no effect of group ($F_{2,76}$ = 1.822, P = 0.169, η^2 = 0.046), CS+ ($F_{1,76}$ = 0.91, P = 0.343, η^2 = 0.012) or their interactions (all Ps > 0.1; **Figure 2B**). To test whether participants across all groups achieved similar levels of extinction, a mixed two-way ANOVA showed no effect of group (30min, 6h and 24h; $F_{2,76}$ = 0.059, P = 0.943, η^2= 0.002), CS+ (CS1+ vs. CS2+; $F_{1,76}$ = 0.258, P = 0.613, η^2 = 0.003) or their interaction ($F_{2,76}$ = 0.504, P = 0.606, η^2 = 0.013) on the last trial of extinction training. Post-hoc t-tests showed that there was no difference between CS+ and CS− responses in all groups on the last trial of extinction (CS1+: t_{26} = −0.397, P = 0.695; CS2+: t_{26} = −0.505, P = 0.618 for the 30min group; CS1+: t_{25} = −0.054, P = 0.957; CS2+: t_{25} = 0.140, P = 0.890 for the 6h group and CS1+: t_{25} = 0.435, P = 0.668; CS2+: t_{25} = −0.317, P = 0.754 for the 24h group; **Figure 2D**).

To assess the effects of the retrieval-extinction procedure as a function of the time delay between fear extinction and test phases, we conducted a 3-way ANOVA with the within-subject factor CS+ (CS1+ vs. CS2+, defined by the mean SCR differences between each CS+ and CS−), time (last trial of extinction vs. first trial of test) and between-subject factor group (30min, 6h and 24h) and found a significant three-way interaction ($F_{2,76}$ = 6.376, P = 0.003, η^2 = 0.144). To disentangle this interaction, we further examined the effects in each group.

In the 30min group, there was no significant effect of time ($F_{1,26}$ = 0.208, P = 0.652, η^2 = 0.008), CS+ ($F_{1,26}$ = 0.888, P = 0.355, η^2 = 0.033) or their interaction (F = 1.039, P = 0.318, ^{2} = 0.038). Post-hoc t-tests showed that the retrieval-extinction training diminished fear responses to both the reminded

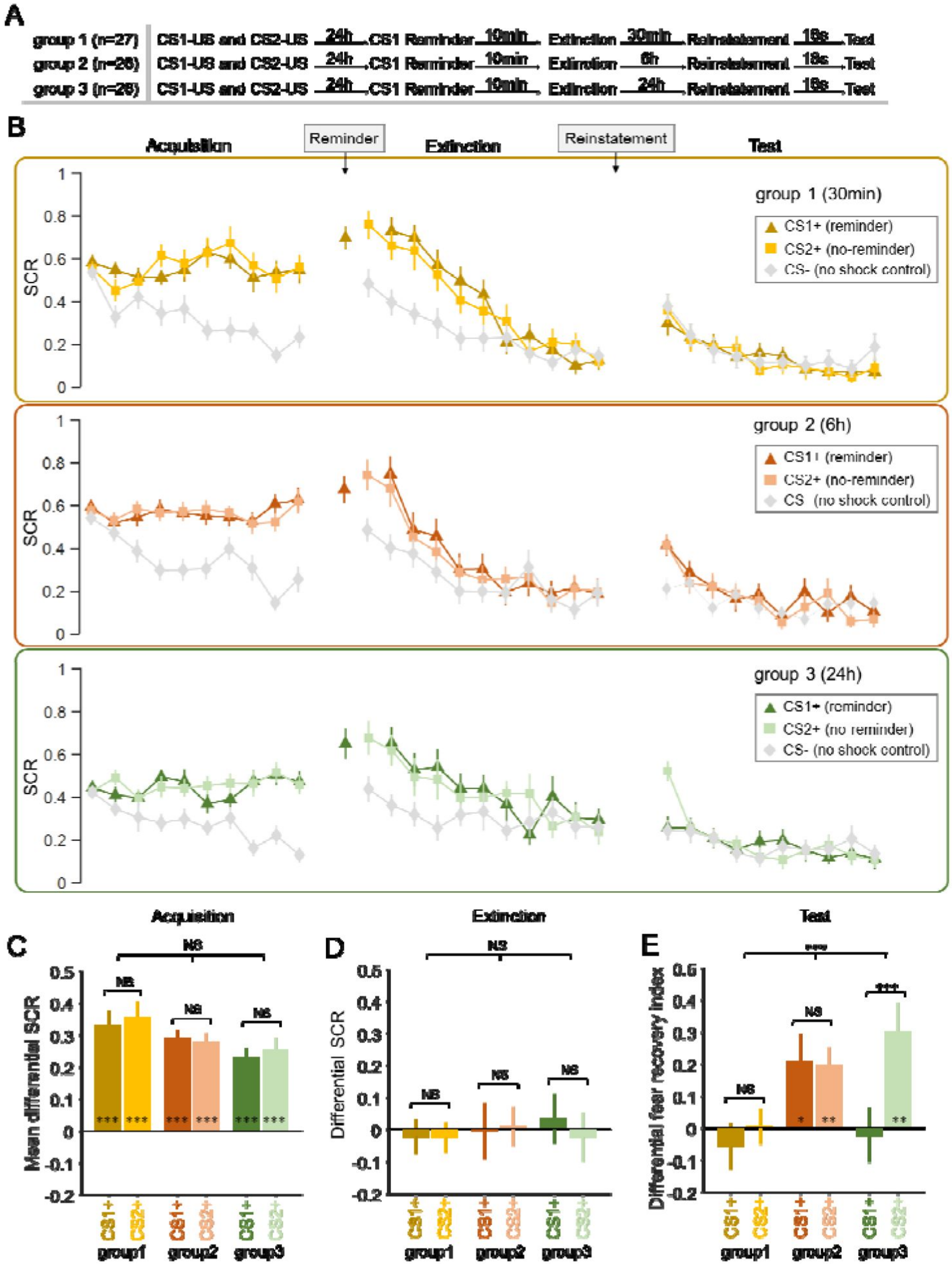


Figure 2|

SCR responses to conditioned stimuli in three groups of study 2 indicating amnesia at different timescales. (**A**) Experimental design and timeline. (**B**) SCRs of fear conditioned stimuli CS1+ (reminder) and CS2+ (No-reminder), and the control stimulus (CS−) across the fear acquisition, extinction and test phases for each group (30min, 6h and 24h groups). (**C**) Mean differential SCRs (CS1+ minus CS− and CS2+ minus CS−) in the acquisition phase (trials in the latter half). (**D**) Differential SCRs in the extinction phase (last trial). (**E**) Differential fear recovery index between CS+ and CS− in the test phase. ****P* < 0.001. ***P* < 0.01. **P* < 0.05. NS: Non-significant. Error bars represent standard errors.

CS+ (differential fear recovery index between CS1+ and CS−, t_{26} = −0.787, P = 0.438; **Figure 2C**) and the non-reminded CS+ (differential fear recovery index between CS2+ and CS−, t_{26} = 0.088, P = 0.93; **Figure 2E**), suggesting a cue-independent short-term amnesia effect. This result indicates that the short-term amnesia effect observed in Study 2 is not reminder-cue specific and can generalize to the non-reminded cues. In addition, there was no significant difference between the fear recovery index associated with CS1+ and CS2+ (t_{26} = −1.019, P = 0.318; **Figure 2E**, also see the supplemental material for direct test of the test phase SCRs).

In contrast to the short-term effect, there was a significant effect of time ($F_{1,25}$ = 9.654, P = 0.005, η^2 = 0.279), but no effect of CS+ ($F_{1,25}$ = 0.037, P = 0.85, η^2= 0.001) or their interaction ($F_{1,25}$ = 0.028, P = 0.869, ^{2} = 0.001) in the 6h (medium-term) group. Post-hoc *t*-tests showed that fear memory recovered for the reminded CS+ (differential fear recovery index between CS1+ and CS−, t_{25} = 2.382, P = 0.025), as well as the non-reminded CS+ (differential fear recovery index between CS2+ and CS−, t_{25} = 3.525, P = 0.002) in the medium-term test of the retrieval-extinction procedure effect, suggesting the failure to suppress the return of extinguished fear memory at such a time scale (**Figure 2E**). There was also no significant difference measured by the fear recovery index between CS1+ and CS2+ (t_{25} = 0.166, P = 0.870; **Figure 2E**).

Finally, in line with previous research on fear memory reconsolidation, significant SCR difference of the fear reinstatement effect between the reminded CS+ and non-reminded CS+ (time × CS+ interaction: $F_{1,25}$ = 16.924, P < 0.001, η^2 = 0.404) was detected in the 24h group via two-way ANOVA. More specifically, the retrieval-extinction procedure only diminished fear responses to the reminded CS+ (differential fear recovery index between CS1+ and CS−, t_{25} = −0.25, P = 0.805), whereas the fear response to the non-reminded CS+ remained significant (differential fear recovery index between CS2+ and CS−, t_{25} = 3.269, P = 0.003; **Figure 2E**). Post-hoc t-test showed that the fear recovery index between CS1+ and CS2+ was significantly different (t_{25} = −4.114, P < 0.001; **Figure 2E**), aligning well with previous literature that suggested the cue-specificity of the fear reconsolidation effect (Lee et al., 2017; Monfils et al., 2009). Similarly, these results remain robust even with the “non-learners” included in the analysis (Figure supplement 2 in the Supplemental Material).

Altogether, in line with our hypothesis, the return of fear memory measured by the reinstatement effect showed distinct patterns at different time scales. The retrieval-extinction procedure diminished fear SCRs for both the reminded and non-reminded CS+, engendering cue-independent amnesia in the short-term test (30min). On the other hand, at the long-term (24h) fear memory test, fear responses were only diminished for the reminded CS+ (but not the non-reminded CS+), consistent with previous literature on memory reconsolidation. Interestingly, SCR measures of the fear memory re-appeared in the medium-term (6h) test and there was no difference between fear responses elicited by the reminded and non-reminded CS+, suggesting that the 6-hour timescale may be at a “limbo” stage where both the immediate and the delayed (reconsolidation) memory update mechanisms are not at work.

Fear amnesia as a function of the thought-control ability

Finally, to assess whether the fear reinstatement effects at different time scale were related to subjects’ thought-control ability, we first ran a one-way ANOVA to confirm that subjects among different groups did not differ in their thought-control abilities measured by the TCAQ scores (F = 0.154, P = 0.857, ^{2} = 0.004). We then ran separate linear regressions between the differential fear recovery index (differential fear recovery index between CS+ and CS−) and thought-control abilities (TCAQ scores) in the three groups. This analysis yielded significant negative correlation between TCAQ score and fear recovery index for both CS+ (CS1+ and CS2+) in the 30min group (**Figure 3A**, main effect of thought-control ability, t_{51} = −3.446, P = 0.003, Bonferroni correction), indicating that subjects endowed with higher thought-control abilities were less likely to experience the return of fear response. However, no significant correlation was observed in the 6h or the 24h group (**Figure 3B** and **C**; Ps > 0.4). Post-hoc analysis of the 30min group showed significant correlation between the fear recovery index of both CS1+ and CS2+ with subjects' thought-control abilities (**Figure 3A**, CS1+: r = −0.461, $P = 0.015$; CS2+: r = −0.413, $P = 0.032$). No significant TCAQ and CS+ interactions were found in the three groups ($Ps > 0.4$).

It is worth noting that correlations between TCAQ scores and fear responses in all three groups were not significant in the late phase of acquisition or the last trial of extinction (all $Ps > 0.10$), indicating that the effect of thought-control ability was specifically reflected in the fear recovery index.

In conclusion, thought-control abilities were associated with fear recovery indices only in the short-term (30min) test for both the reminded and non-reminded CS+ but not at longer time scales (6h and 24h). The 30min group results are consistent with previous research which suggested that people with better capability to resist intrusive thoughts also performed better in motivated forgetting in both declarative and associative memories (Küpper et al., 2014; Wang et al., 2021).

dlPFC dependent short-term fear memory amnesia

In study 3 (**Figure 4A**), we examined whether intact activity of the right dlPFC (rdlPFC) was also necessary for short-term fear amnesia, in addition to the memory retrieval cue (reminder). To test this hypothesis, we set up a reminder CS+ (CS1+ vs. CS2+) x reminder (reminder vs. no-reminder) x cTBS (rdlPFC vs. vertex) x trial number four-way ANOVA against the differential SCRs (CS+ minus CS−) for the acquisition and extinction phases. In the acquisition phase, as expected, there were no significant main effects of reminder ($F_{1,71} = 1.291$, $P = 0.260$, $\eta^2 = 0.018$), cTBS ($F_{1,71} < 0.001$, $P = 0.990$, $\eta^2 < 0.001$), CS+ ($F_{1,71} = 1.927$, $P = 0.169$, $\eta^2 = 0.026$) or their interactions (all $Ps > 0.1$; **Figure 4B** and **C**) but a significant main effect of trial number ($F_{9,639} = 8.054$, $P < 0.001$, $\eta^2 = 0.102$), indicating successful learning of CS+ stimuli. Indeed, post-hoc tests confirmed that the SCRs of CS+ was significantly higher than that of the CS− on the late phase (last 5 trials) of acquisition in all four groups (CS1+: $t_{18} = 7.735$, $P < 0.001$; CS2+: $t_{18} = 8.546$, $P < 0.001$ for the R-PFC group; CS1+: $t_{17} = 8.568$, $P < 0.001$; CS2+: $t_{17} = 8.478$, $P < 0.001$ for the R-VER group; CS1+: $t_{17} = 6.032$, $P < 0.001$; CS2+: $t_{17} = 7.722$, $P < 0.001$ for the NR-PFC group and CS1+: $t_{19} = 9.110$, $P < 0.001$; CS2+: $t_{19} = 6.538$, $P < 0.001$ for the NR-VER group, **Figure 4C**).

A similar mixed-effect four-way ANOVA was also performed on the SCRs in the extinction phase, which showed significant effects of trial ($F_{10,710} = 8.567$, $P < 0.001$, $^2 = 0.108$), cTBS ($F_{1,71} = 6.269$, $P = 0.015$, $\eta^2 = 0.081$) and cTBS × trial interaction ($F_{10,710} = 2.292$, $P = 0.030$, $\eta^2 = 0.031$), but no effect of reminder ($F_{1,71} = 1.894$, $P = 0.173$, $\eta^2 = 0.026$), CS+ ($F_{1,71} = 0.398$, $P = 0.530$, $\eta^2 = 0.006$) or other interactions (all $Ps > 0.1$; **Figure 4B**). To test whether participants across all groups achieved similar levels of extinction, a mixed-effect three-way ANOVA was performed on the last trial of extinction training and there was no significant effect of reminder (reminder and no-reminder; $F_{1,71} = 3.112$, $P = 0.082$, $\eta^2 = 0.042$), cTBS ($F_{1,71} = 0.929$, $P = 0.338$, $\eta^2 = 0.013$), CS+ (CS1+ vs. CS2+; $F_{1,71} = 1.623$, $P = 0.207$, $\eta^2 = 0.022$) or their interactions (all $Ps > 0.1$; **Figure 4D**). Furthermore, post-hoc *t*-tests showed that there was no difference between CS+ and CS− responses in all groups on the last trial of extinction (CS1+: $t_{18} = 1.310$, $P = 0.207$; CS2+: $t_{18} = 1.298$, $P = 0.211$ for the R-PFC group; CS1+: $t_{17} = 0.839$, $P = 0.413$; CS2+: $t_{17} = 1.090$, $P = 0.291$ for the R-VER group; CS1+: $t_{17} = 0.380$, $P = 0.709$; CS2+: $t_{17} = 0.624$, $P = 0.541$ for the NR-PFC group and CS1+: $t_{19} = -1.434$, $P = 0.168$; CS2+: $t_{19} = -0.535$, $P = 0.599$ for the NR-VER group; **Figure 4D**).

To assess the effects of the right dlPFC on the memory retrieval related short-term amnesia, we conducted a mixed-effect 3-way ANOVA with the within-subject factors CS+ (CS1+ vs. CS2+), between-subject factors reminder (reminder vs. no-reminder) and cTBS (dlPFC vs. vertex) on fear recovery index (the difference between the first test trial and the last extinction trial for CS1+ and CS2+), which revealed a significant effect of reminder ($F_{1,71} = 8.920$, $P = 0.004$, $\eta^2 = 0.112$), cTBS ($F_{1,71} = 7.180$, $P = 0.009$, $\eta^2 = 0.092$) and reminder × cTBS interaction ($F_{1,71} = 10.613$, $P = 0.002$, $\eta^2 = 0.130$), but no effect of CS+ ($F_{1,71} = 0.518$, $P = 0.474$, $\eta^2 = 0.007$) or other interactions (all $Ps > 0.1$;

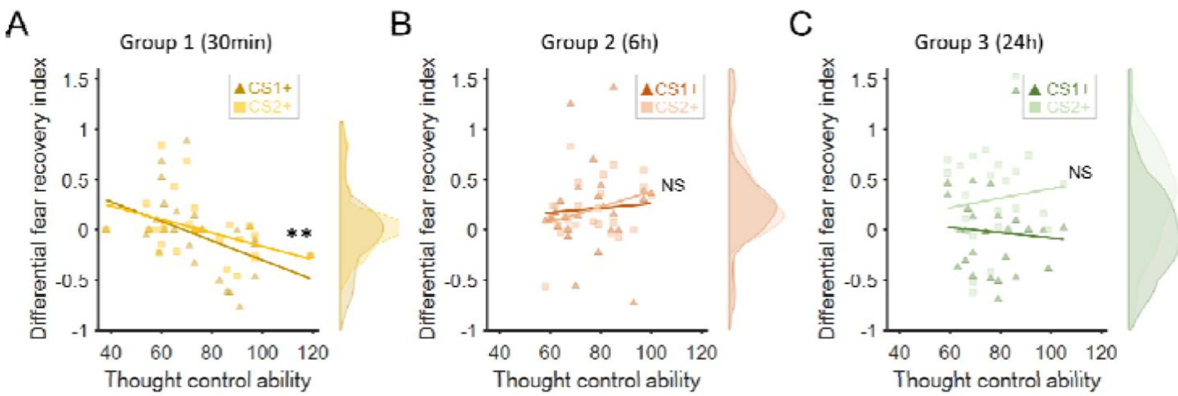


Figure 3|

Fear recovery as a function of thought-control abilities. (**A**)Thought control ability was significantly correlated with fear recovery index in the 30min group (P = 0.003, Bonferroni correction), but not in the 6h (**B**) or 24h group (**C**, Ps > 0.7). The violin graphs indicate the distribution of fear recovery index across subjects (**Figure 2E**). **P < 0.01. NS: Non-significant.

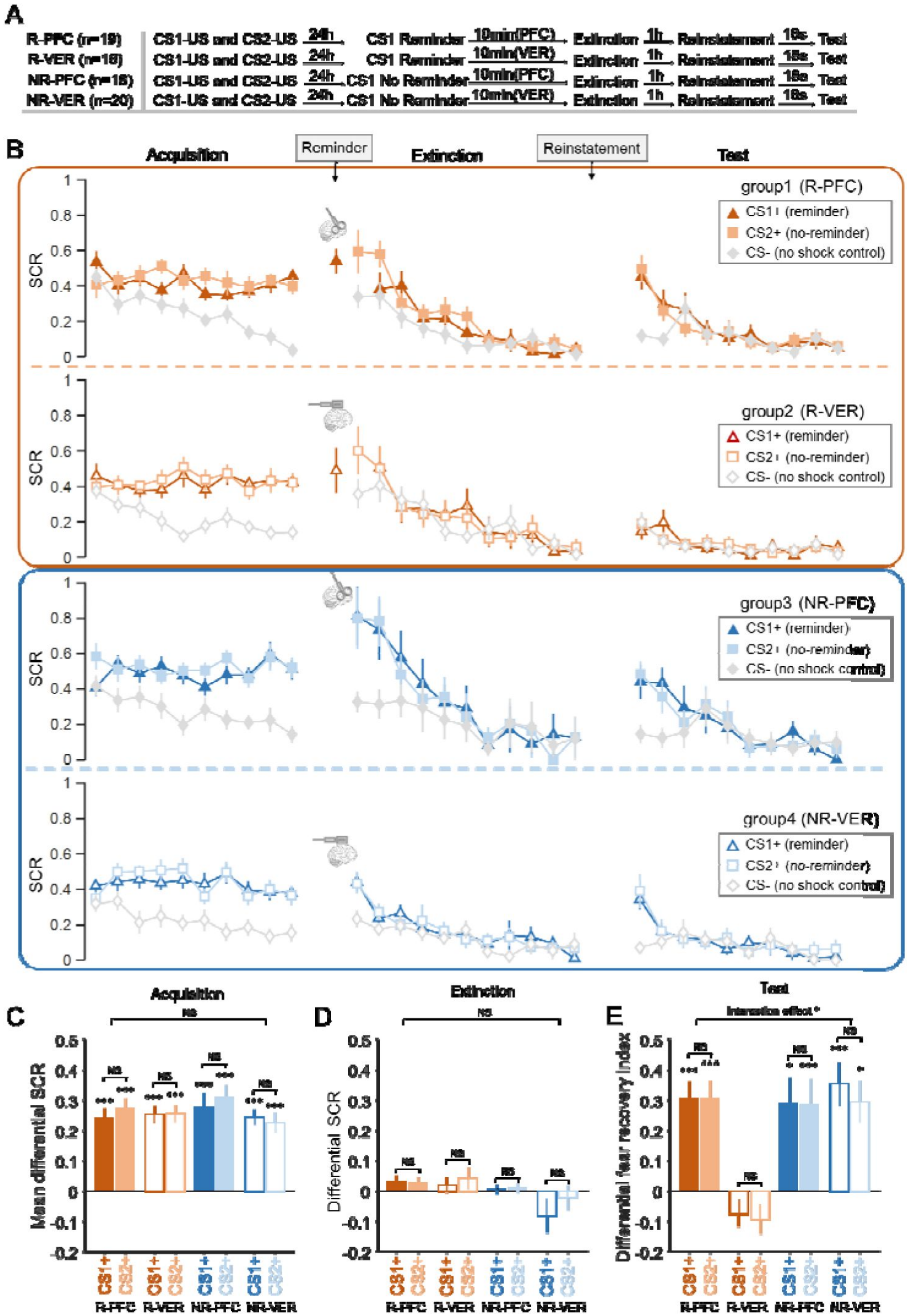


Figure 4|

SCR responses to conditioned stimuli in the cTBS study (study 3) indicating that both intact dlPFC and cue reminder are required for short-term memory amnesia. (**A**) Experimental design and timeline. (**B**) SCRs of fear conditioned stimuli CS1+ (reminder) and CS2+ (No-reminder), and the control stimulus (CS−) across the fear acquisition, extinction and test phases for each group (R-PFC, R-VER, NR-PFC and NR-VER groups). (**C**) Mean differential SCRs (CS1+ minus CS− and CS2+ minus CS−) in the acquisition phase (trials in the latter half). (**D**) Differential SCRs in the extinction phase (last trial). **(E)** Differential fear recovery index between CS+ and CS− in the test phase. ****P* < 0.001. **P* < 0.05. NS: Non-significant. Error bars represent standard errors.

Figure 4C). Post-hoc *t*-tests showed that there were significant fear reinstatement responses in the R-PFC group (CS1+: t_{18} = 5.182, P < 0.001; CS2+: t_{18} = 5.258, P < 0.001), the NR-PFC group (CS1+: t_{17} = 3.496, P = 0.003; CS2+: t_{17} = 4.628, P < 0.001) and the NR-VER group (CS1+: t_{19} = 4.885, P < 0.001; CS2+: t_{19} = 3.262, P = 0.004), but not in the R-VER group (CS1+: t_{17} = −1.640, P = 0.119; CS2+: t_{17} = −0.744, P = 0.467) (**Figure 4E**), suggesting that both memory retrieval and intact dlPFC function are necessary for the short-term fear amnesia.

Finally, we went on to examine whether the fear reinstatement response was related to the thought-control ability. A two-way ANOVA of reminder (reminder vs. no-reminder) x cTBS (rdlPFC vs. vertex) confirmed that there were no significant effects of reminder ($F_{1,71}$ = 0.366, P = 0.547, η^2 = 0.005), cTBS ($F_{1,71}$ = 0.606, P = 0.439, η^2 = 0.008) or their interaction ($F_{1,71}$ = 1.216, P = 0.274, η^2 = 0.017) on the TCAQ scores. We then conducted simple linear regressions on different groups (as in study 2) and found a significant correlation between the thought-control ability and differential fear recovery index in the R-VER group (**Figure 5B**, t_{33} = −3.283, P = 0.008, Bonferroni correction across 4 groups), consistent with the results of the short term (30min) amnesia effect in study 2 (**Figure 3A**). Further post-hoc analysis of the individual CS+ showed that the correlations between thought-control ability and fear recovery index were significant for both CS+ (CS1+: r = −0.557, P = 0.016; CS2+: r = −0.483, P = 0.042). However, there was no such correlation in the R-PFC group (**Figure 5A**, t_{35}= 0.719, P = 0.477), the NR-PFC group (**Figure 5C**, t_{33} = −1.334, P = 0.192) or the NR-VER group (**Figure 5D**, t_{37} = −0.025, P = 0.980) and the interaction effects of thought-control ability and CS+ (CS1+ and CS2+) were not significant in all 4 groups (*P*s > 0.3). Importantly, there was no significant correlation between thought-control ability and SCR in the late phase of acquisition or the last trial of extinction in four groups (all *Ps* > 0.10), suggesting a specific relationship between thought-control ability and short-term fear recovery.

Discussion

Our results provide direct evidence to support the hypothesis that the memory reactivation might engage distinct cognitive mechanisms for fear memory modulation, which can be separated by their temporal dynamics, cue specificity and the dependence of thought-control ability. In line with previous research on fear memory reconsolidation (Liu et al., 2014), our research replicated the reported results that the retrieval-extinction paradigm blocked fear responses to the reminded CS+ while leaving non-reminded CS+ response intact (cue-specificity) in the long-term test (**Figure 6A** and **B**). However, thought-control ability was not related to the long-term fear amnesia (**Figure 6C**). Therefore, findings of the short-term fear amnesia suggest that the reconsolidation framework falls short to accommodate this more immediate effect (**Figure 6A** and **B**).

Research in both declarative and associative emotional memories suggest that memory retrieval is not a simple act of obtaining a copy of stored information, but provides a critical time window where the old memory is eligible to be integrated with new information (Gershman, Schapiro, et al., 2013; Hupbach et al., 2007; Monfils et al., 2009). Research in the field of declarative memory has shown that the motivated forgetting effect can appear early (10-30 minutes) after the memory suppression practices such as the influential think/no-think (TNT) paradigm (Anderson and Green, 2001), suggesting that memory retrieval may also facilitate short-term memory update (Benoit and Anderson, 2012; Engen and Anderson, 2018). Unlike the slow protein synthesis process involved in memory reconsolidation, the motivated forgetting effect was accompanied by heightened dorsolateral prefrontal cortex (dlPFC) activity, diminished hippocampal activation and altered functional connectivity between these brain regions and hence more adaptive (Anderson et al., 2004; Wimber et al., 2015).

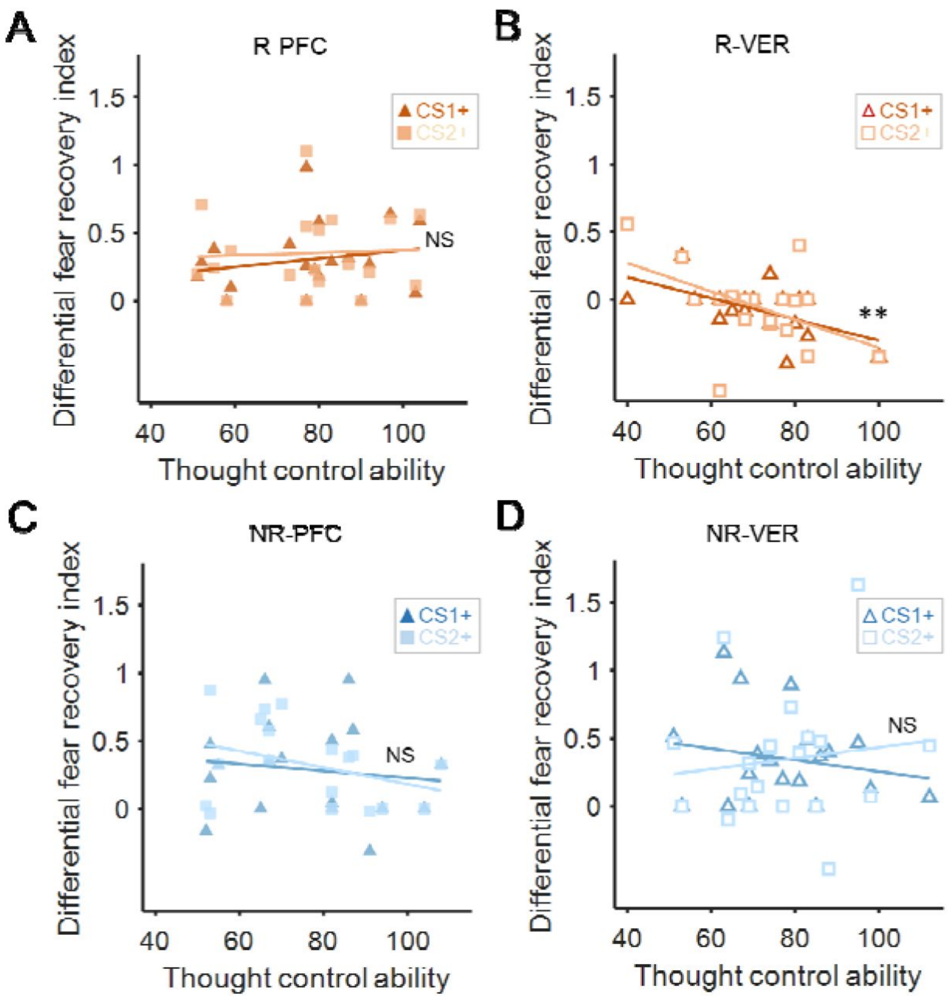


Figure 5|

The correlation between differential fear recovery index and thought-control ability in four cTBS groups. (**A**) There was no significant correlation between differential fear recovery index and thought-control abilities in the R-PFC group ($P > 0.4$). (**B**) However, in the R-VER group, the correlation between thought control ability scores and the differential fear recovery index was significant ($P = 0.008$, Bonferroni correction), with high thought-control ability participants showing less fear recovery for both CSs+. Such correlation was not observed in the (**C**) NR-PFC or the (**D**) NR-VER group (*Ps* >0.4). ** $P < 0.01$, NS: Non-significant.

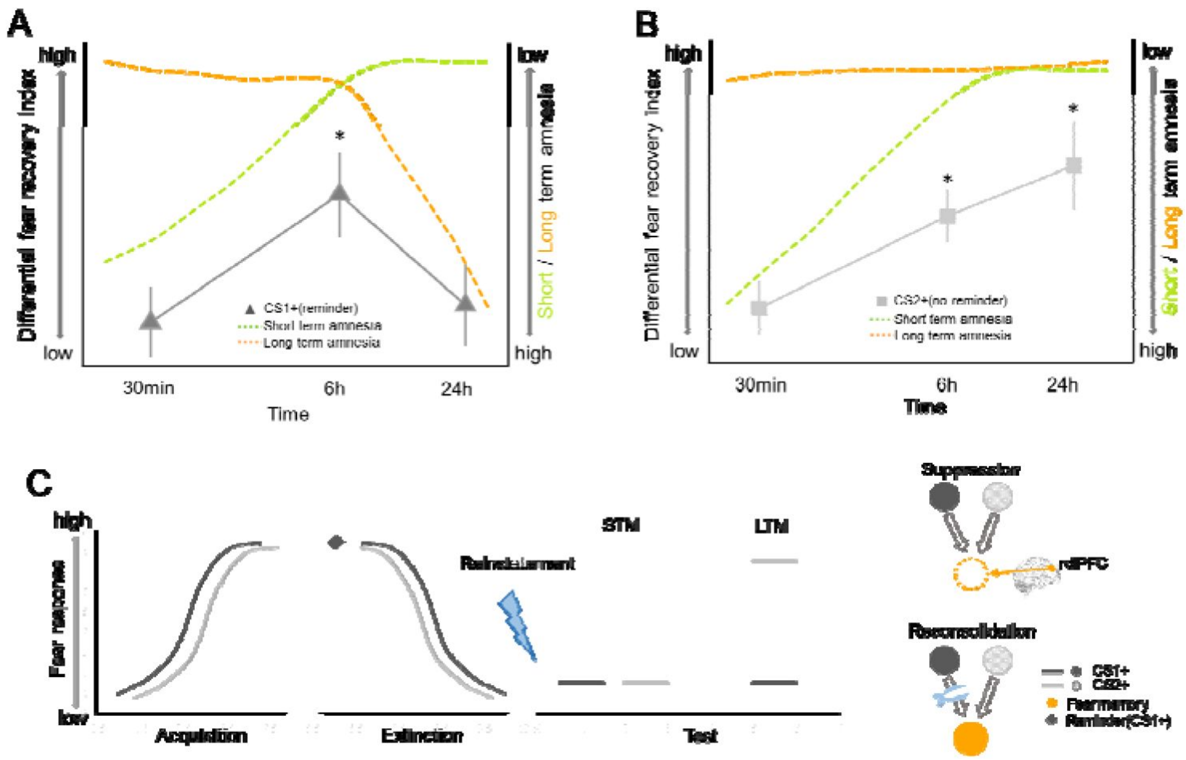


Figure 6|

Time courses of short- and long-term amnesia. (**A**) At the short interval (30min to 1h), fear recovery of the reminded CS (CS1+) is inhibited (green). As time progresses (from 6h to 24h), amnesia effect is likely due to the fear memory reconsolidation effect emerged later (orange). Actual SCR data in black solid line. (**B**) Fear amnesia of the non-reminded CS (CS2+) is only evident at the 30-min interval and such effect starts to decay as test interval increases (green). However, long-term amnesia does not generalize to CS2+ (orange, cue-dependence). The observed SCR data (black solid line) paralleled the prediction from both the short-term and the long-term effects as the interval length increased (from 30min to 24h). (**C**) Schematics depicting the effect of cue-reminder on fear memory retention. After both CS1+ (black) and CS2+ (grey) successfully elicit fear responses in the acquisition phase, CS1+ is reminded (black) before both CS+ go through the extinction training. The lack of both CS1+ and CS2+ fear responses in the short-term memory (STM) test (30 min) might be explained by the dlPFC dependent direct suppression effect (dotted circle of US representation). On the other hand, the cue-specific fear amnesia effect in the long-term memory (LTM) test (24h) of CS1+ but not CS2+ could be attributed to the reconsolidation effect specific to CS1+. * $P < 0.05$, Error bars represent standard errors.

Previous studies indicate that a suppression mechanism can be characterized by three distinct features: first, the memory suppression effect tends to emerge early, usually 10-30 mins after memory suppression practice and can be transient (MacLeod and Macrae, 2001; Saunders and MacLeod, 2002); second, the memory suppression practice seems to directly act upon the unwanted memory itself (Levy and Anderson, 2002), such that the presentation of other cues originally associated with the unwanted memory also fails in memory recall (cue-independence); third, the magnitude of memory suppression effects is associated with individual difference in control abilities over intrusive thoughts (Küpper et al., 2014). The short-term fear amnesia effects we identified are fundamentally different from memory reconsolidation effects and fit closely with the characteristics of the memory suppression effect. Inspired by the similarities between our results and suppression-induced declarative memory amnesia (Gagnepain et al., 2017), we speculate that the retrieval-extinction procedure might facilitate a spontaneous memory suppression process and thus yield a short-term amnesia effect. Accordingly, the activated fear memory induced by the retrieval cue would be subjected to an automatic fear memory suppression through the extinction training (Anderson and Floresco, 2022).

In our experiments, subjects were not explicitly instructed to suppress their fear expression, yet the retrieval-extinction training significantly decreased short-term fear expression. These results are consistent with the short-term amnesia induced with the more explicit suppression intervention (Anderson et al., 1994; Kindt and Soeter, 2018; Speer et al., 2021; Wang et al., 2021; Wells and Davies, 1994). It is worth noting that although consciously repelling unwanted memory is a standard approach in memory suppression paradigm, it is possible that the engagement of the suppression mechanism can be unconscious. For example, in the retrieval-induced forgetting (RIF) paradigm, recall of a stored memory impairs the retention of related target memory and this forgetting effect emerges as early as 20 minutes after the retrieval procedure, suggesting memory suppression or inhibition can occur in a more spontaneous and automatic manner (Imai et al., 2014). Moreover, subjects with trauma histories exhibited more suppression-induced forgetting for both negative and neutral memories than those with little or no trauma (Hulbert and Anderson, 2018). Similarly, people with higher self-reported thought-control capabilities showed more severe cue-independent memory recall deficit, suggesting that suppression mechanism is associated with individual differences in spontaneous control abilities over intrusive thoughts (Küpper et al., 2014). It has also been suggested that similar automatic mechanisms might be involved in organic retrograde amnesia of traumatic childhood memories (Schacter et al., 2012; Schacter et al., 1996).

Therefore, it is theoretically possible that similar brain mechanisms, such as the dynamic brain activity and connectivity modulation, involved in memory suppression may be deployed spontaneously and act as a general cognitive mechanism to cope with intrusive memories in daily life. Indeed, recently researchers have proposed that the standard retrieval-extinction paradigm to study fear memory might also engage automatic memory suppression (Anderson and Floresco, 2022). Consistent with this hypothesis, we showed that the retrieval cue was necessary to produce the short-term amnesia effect (**Figure 1**) and critically, our results fitted well with the three key characteristics of the active suppression mechanism: temporal dynamics, cue-specificity and the dependence of thought-control ability (**Figures 2** and **3**).

To calibrate the temporal boundaries of the short and long-term effects, we also included a separate group (medium-term group) where fear expression was tested 6 hours after the retrieval-extinction training. Interestingly, fear returned at this time point (**Figure 6A** and **B**), confirming the 6-hour lower time boundary for the hypothesized memory reconsolidation in previous animal and human studies (Agren, 2014; Agren et al., 2012; Debiec et al., 2002; Kindt and Soeter, 2018; Lee, 2008; Nader, Schafe, & Le Doux, 2000; Speer et al., 2021). However, the return of fear expression in the medium-term group also indicates that the short-term fear amnesia is transient, laying out the upper time boundary for the hypothesized fear memory suppression effect. Although mixed results have been reported regarding the durability of suppression effects in the declarative memory studies (Meier et al., 2011; Storm et al., 2012), future research will be needed to investigate whether the short-term effect we observed is specifically related to associative memory or the spontaneous nature of suppression (**Figure 6C**). Previous research also showed that separate mnemonic processes can be involved after emotional memory retrieval. For example, by varying the duration length and repetition number of memory cue re-exposure, different levels of memory persistence (reconsolidation, extinction or a "limbo" state in between) can be induced, suggesting a neural titration mechanism for memory de-stabilization and re-stabilization (Faliagkas et al., 2018; Merlo et al., 2014). These works identified important "boundary conditions" of memory retrieval in affecting the retention of the maladaptive emotional memories. In our study, however, we showed that even within a boundary condition previously thought to elicit memory reconsolidation, mnemonic processes other than reconsolidation could also be at work, and these processes jointly shape the persistence of fear memory.

Our results also corroborate with a recent study showing that blocking fear reconsolidation using post-retrieval repetitive transcranial magnetic stimulation (rTMS) did not influence the short-term fear memory expression (Borgomaneri et al., 2020b). However, as we demonstrated in studies 2 & 3, memory retrieval and intact rdlPFC function were both necessary for the short-term fear amnesia after the extinction training. Therefore, instead of "intact short-term fear expression by the rTMS over the dlPFC", their results could also be interpreted such that the rTMS on the dlPFC demolished the otherwise short-term fear amnesia induced by the memory retrieval cue. Indeed, a series of neuroimaging studies have revealed that the prefrontal neural network including anterior cingulate cortex, anterior ventrolateral prefrontal cortex and the dlPFC is directly engaged in purging the unwanted declarative memory from consciousness (Anderson et al., 2004; Benoit and Anderson, 2012; Levy and Anderson, 2002). Moreover, previous research has established a causal relationship between dlPFC activity and the memory suppression mechanism underlying retrieval-induced forgetting using transcranial direct current stimulation (tDCS) (Penolazzi et al., 2014; Stramaccia et al., 2017; Valle et al., 2020). In our experiment (studies 2 and 3), when the dlPFC function was intact, subjects' thought-control abilities were positively correlated with fear amnesia in the test phase. Such correlations disappeared in the no-reminder groups (NR-VER and NR-PFC) and the R-PFC group (**Figures 3** and **5**), establishing a causal link between the dlPFC activity and short-term fear amnesia.

It should be noted that while our long-term amnesia results were consistent with the fear memory reconsolidation literatures, there were also studies that failed to observe fear prevention (Chalkia, Schroyens, et al., 2020; Chalkia, Van Oudenhove, & Beckers, 2020; Schroyens et al., 2023). Although the memory reconsolidation framework provides a viable explanation for the long-term amnesia, more evidence is required to validate the presence of reconsolidation, especially at the neurobiological level (Elsey et al., 2018). While it is beyond the scope of the current study to discuss the discrepancies between these studies, one possibility to reconcile these results concerns the procedure for the retrieval-extinction training. It has been shown that the eligibility for old memory to be updated is contingent on whether the old memory and new observations can be inferred to have been generated by the same latent cause (Gershman et al., 2017; Gershman and Niv, 2012). For example, prevention of the return of fear memory can be achieved through gradual extinction paradigm, which is thought to reduce the size of prediction errors to inhibit the formation of new latent causes (Gershman, Jones, et al., 2013). Therefore, the effectiveness of the retrieval-extinction paradigm might depend on the reliability of such paradigm in inferring the same underlying latent cause. Furthermore, other studies highlighted the importance of memory storage *per se* and suggested that memory retention was encoded in the memory engram cell ensemble connectivity whereas the engram cell synaptic plasticity is crucial for memory retrieval (Ryan et al., 2015; Tonegawa, Liu, et al., 2015; Tonegawa, Pignatelli, et al., 2015). It remains to be tested how the cue-independent short-term and cue-dependent long-term amnesia effects we observed could correspond to the engram cell synaptic plasticity and functional connectivity among engram cell ensembles (**Figure 6**). This is particularly important, since the cue-independent characteristic of the short-term amnesia suggest that either different memory cues fail to evoke engram cell activities, or the retrieval-extinction training transiently inhibits connectivity among engram cell ensembles. Finally, SCR is only one aspect of the fear expression, how the retrieval-extinction paradigm might affect subjects' other emotional (such as the startle response) and cognitive fear expressions such as reported fear expectancy needs to be tested in future studies since they do not always align with each other (Kindt et al., 2009; Sevenster et al., 2012, 2013).

These results help shed light on the dynamics of memory modulation after memory activation and designing novel treatment of psychiatric disorders caused by excessive fear or anxiety (Chen et al., 2021). Memory reconsolidation and suppression both have been studied thoroughly in separate fields. Specifically, aversive memories are usually associated with different retrieval cues, and interfering memory reconsolidation only blocks fear memory recall from the retrieval memory probe but not the others (Debiec et al., 2002; Kindt and Soeter, 2018; Lee, 2008; Nader, Schafe, & Le Doux, 2000; Speer et al., 2021). On the other hand, memory suppression is cue-independent and directly suppresses the fear memory trace yet only effective with limited durability (MacLeod and Macrae, 2001). Although future research is clearly needed to understand the brain mechanisms for the short-term amnesia and its potential connection to the memory suppression effect, especially in relevance to clinical populations such as posttraumatic stress disorder (PTSD) and phobia patients (Homan et al., 2019; Stramaccia et al., 2021), our results provide a general framework to highlight the potential of memory retrieval to triggering different memory updating mechanisms with unique temporal dynamics.

Data availability

All the data of this study are available for download at *https://osf.io/9agvk/*.

Additional information

Transparency

This work was supported by the National Science and Technology Innovation 2030 Major Program (2021ZD0203702), National Natural Science Foundation of China Grants (32071090, 31871140) to J.L.

The authors have declared no conflict of interests.

Additional files

Supplemental file

eLife

References

Agren T. 2014) **Human reconsolidation: a reactivation and update** *Brain Res Bull* **105**:70–82 https://doi.org/10.1016/j.brainresbull.2013.12.010

Agren T., Engman J., Frick A., Björkstrand J., Larsson E. M., Furmark T., Fredrikson M. 2012) **Disruption of reconsolidation erases a fear memory trace in the human amygdala** *Science* **337**:1550–1552 https://doi.org/10.1126/science.1223006

Alberini C. M. 2005) **Mechanisms of memory stabilization: are consolidation and reconsolidation similar or distinct processes?** *Trends Neurosci* **28**:51–56 https://doi.org/10.1016/j.tins.2004.11.001

Anderson M. C., Bjork E. L., Bjork R. A. 2000) **Retrieval-induced forgetting: evidence for a recall-specific mechanism** *Psychon Bull Rev* **7**:522–530 https://doi.org/10.3758/bf03214366

Anderson M. C., Bjork R. A., Bjork E. L. 1994) **Remembering can cause forgetting: retrieval dynamics in long-term memory** *Journal of experimental psychology. Learning, memory, and cognition* **20**:1063–1087 https://doi.org/10.1037//0278-7393.20.5.1063

Anderson M. C., Floresco S. B. 2022) **Prefrontal-hippocampal interactions supporting the extinction of emotional memories: the retrieval stopping model** *Neuropsychopharmacology* **47**:180–195 https://doi.org/10.1038/s41386-021-01131-1

Anderson M. C., Green C. 2001) **Suppressing unwanted memories by executive control** *Nature* **410**:366–369 https://doi.org/10.1038/35066572

Anderson M. C., Ochsner K. N., Kuhl B., Cooper J., Robertson E., Gabrieli S. W., Glover G. H., Gabrieli J. D. 2004) **Neural systems underlying the suppression of unwanted memories** *Science* **303**:232–235 https://doi.org/10.1126/science.1089504

Benoit R. G., Anderson M. C. 2012) **Opposing mechanisms support the voluntary forgetting of unwanted memories** *Neuron* **76**:450–460 https://doi.org/10.1016/j.neuron.2012.07.025

Borgomaneri S., Battaglia S., Garofalo S., Tortora F., Avenanti A., di Pellegrino G. 2020a) **State-Dependent TMS over Prefrontal Cortex Disrupts Fear-Memory Reconsolidation and Prevents the Return of Fear** *Curr Biol* **30**:3672–3679 https://doi.org/10.1016/j.cub.2020.06.091

Borgomaneri S., Battaglia S., Garofalo S., Tortora F., Avenanti A., di Pellegrino G. 2020b) **State-Dependent TMS over Prefrontal Cortex Disrupts Fear-Memory Reconsolidation and Prevents the Return of Fear** *Curr Biol* **30**:3672–3679 https://doi.org/10.1016/j.cub.2020.06.091

Censor N., Dimyan M. A., Cohen L. G. 2010) **Modification of existing human motor memories is enabled by primary cortical processing during memory reactivation** *Curr Biol* **20**:1545–1549 https://doi.org/10.1016/j.cub.2010.07.047

Chalkia A., Schroyens N., Leng L., Vanhasbroeck N., Zenses A.-K., Van Oudenhove L., Beckers T. 2020) **No persistent attenuation of fear memories in humans: A registered replication of the reactivation-extinction effect** *Cortex* **129**:496–509 https://doi.org/10.1016/j.cortex.2020.04.017

Chalkia A., Van Oudenhove L., Beckers T. 2020) **Preventing the return of fear in humans using reconsolidation update mechanisms: A verification report of Schiller, et al. (2010)** *Cortex* **129**:510–525 https://doi.org/10.1016/j.cortex.2020.03.031

Chen W., Li J., Zhang X., Dong Y., Shi P., Luo P., Zheng X. 2021) **Retrieval-extinction as a reconsolidation-based treatment for emotional disorders:Evidence from an extinction retention test shortly after intervention** *Behav Res Ther* **139**:103831 https://doi.org/10.1016/j.brat.2021.103831

de Voogd L. D., Phelps E. A. 2020) **A cognitively demanding working-memory intervention enhances extinction Research article** *Scientific report* **10**:7020 https://doi.org/10.1038/s41598-020-63811-0

Debiec J., LeDoux J. E., Nader K. 2002) **Cellular and systems reconsolidation in the hippocampus** *Neuron* **36**:527–538 https://doi.org/10.1016/s0896-6273(02)01001-2

Dudai Y. 2006) **Reconsolidation: the advantage of being refocused** *Curr Opin Neurobiol* **16**:174–178 https://doi.org/10.1016/j.conb.2006.03.010

Dunsmoor J. E., Campese V. D., Ceceli A. O., LeDoux J. E., Phelps E. A. 2015) **Novelty-facilitated extinction: providing a novel outcome in place of an expected threat diminishes recovery of defensive responses** *Biological psychiatry* **78**:203–209 https://doi.org/10.1016/j.biopsych.2014.12.008

Dunsmoor J. E., Otto A. R., Phelps E. A. 2017) **Stress promotes generalization of older but not recent threat memories** *Proceedings of the National Academy of Sciences of the United States of America* **114**:9218–9223 https://doi.org/10.1073/pnas.1704428114

Elsey J. W. B., Van Ast V. A., Kindt M. 2018) **Human memory reconsolidation: A guiding framework and critical review of the evidence** *Psychological bulletin* **144**:797–848 https://doi.org/10.1037/bul0000152

Engen H. G., Anderson M. C. 2018) **Memory Control: A Fundamental Mechanism of Emotion Regulation** *Trends in Cognitive Sciences* **22**:982–995 https://doi.org/10.1016/j.tics.2018.07.015

Faliagkas L., Rao-Ruiz P., Kindt M. 2018) **Emotional memory expression is misleading: delineating transitions between memory processes** *Current Opinion in Behavioral Sciences* **19**:116–122 https://doi.org/10.1016/j.cobeha.2017.12.018

Faul F., Erdfelder E., Buchner A., Lang A.-G. 2009) **Statistical power analyses using G*Power 3.1: Tests for correlation and regression analyses** *Behavior Research Methods* **41**:1149–1160 https://doi.org/10.3758/BRM.41.4.1149

Gagnepain P., Hulbert J., Anderson M. C. 2017) **Parallel Regulation of Memory and Emotion Supports the Suppression of Intrusive Memories** *The Journal of neuroscience* **37**:6423–6441 https://doi.org/10.1523/jneurosci.2732-16.2017

Gershman S. J., Jones C. E., Norman K. A., Monfils M. H., Niv Y. 2013) **Gradual extinction prevents the return of fear: implications for the discovery of state** *Front Behav Neurosci* **7**:164 https://doi.org/10.3389/fnbeh.2013.00164

Gershman S. J., Monfils M. H., Norman K. A., Niv Y. 2017) **The computational nature of memory modification** *eLife* **6** https://doi.org/10.7554/eLife.23763

Gershman S. J., Niv Y. 2012) **Exploring a latent cause theory of classical conditioning** *Learn Behav* **40**:255–268 https://doi.org/10.3758/s13420-012-0080-8

Gershman S. J., Schapiro A. C., Hupbach A., Norman K. A. 2013) **Neural context reinstatement predicts memory misattribution** *The Journal of neuroscience* **33**:8590–8595 https://doi.org/10.1523/jneurosci.0096-13.2013

Heo M., Leon A. C. 2010) **Sample sizes required to detect two-way and three-way interactions involving slope differences in mixed-effects linear models** *J Biopharm Stat* **20**:787–802 https://doi.org/10.1080/10543401003618819

Homan P., Levy I., Feltham E., Gordon C., Hu J., Li J., Pietrzak R. H., Southwick S., Krystal J. H., Harpaz-Rotem I., Schiller D. 2019) **Neural computations of threat in the aftermath of combat trauma** *Nature neuroscience* **22**:470–476 https://doi.org/10.1038/s41593-018-0315-x

Hu J., Wang W., Homan P., Wang P., Zheng X., Schiller D. 2018) **Reminder duration determines threat memory modification in humans** *Scientific report* **8**:8848 https://doi.org/10.1038/s41598-018-27252-0

Hulbert J. C., Anderson M. C. 2018) **What doesn't kill you makes you stronger: Psychological trauma and its relationship to enhanced memory control** *Journal of experimental psychology. General* **147**:1931–1949 https://doi.org/10.1037/xge0000461

Hupbach A., Gomez R., Hardt O., Nadel L. 2007) **Reconsolidation of episodic memories: a subtle reminder triggers integration of new information** *Learn Mem* **14**:47–53 https://doi.org/10.1101/lm.365707

Imai H., Kim D., Sasaki Y., Watanabe T. 2014) **Reward eliminates retrieval-induced forgetting** *Proceedings of the National Academy of Sciences of the United States of America* **111**:17326–17329 https://doi.org/10.1073/pnas.1404469111

Kindt M., Soeter M. 2018) **Pharmacologically induced amnesia for learned fear is time and sleep dependent** *Nature communications* **9**:1316 https://doi.org/10.1038/s41467-018-03659-1

Kindt M., Soeter M., Vervliet B. 2009) **Beyond extinction: erasing human fear responses and preventing the return of fear** *Nature neuroscience* **12**:256–258 https://doi.org/10.1038/nn.2271

Küpper C. S., Benoit R. G., Dalgleish T., Anderson M. C. 2014) **Direct suppression as a mechanism for controlling unpleasant memories in daily life** *Journal of experimental psychology. General* **143**:1443–1449 https://doi.org/10.1037/a0036518

Lee J. L. 2008) **Memory reconsolidation mediates the strengthening of memories by additional learning** *Nature neuroscience* **11**:1264–1266 https://doi.org/10.1038/nn.2205

Lee J. L. C., Nader K., Schiller D. 2017) **An Update on Memory Reconsolidation Updating** *Trends in Cognitive Sciences* **21**:531–545 https://doi.org/10.1016/j.tics.2017.04.006

Levy B. J., Anderson M. C. 2002) **Inhibitory processes and the control of memory retrieval** *Trends in Cognitive Sciences* **6**:299–305 https://doi.org/10.1016/s1364-6613(02)01923-x

Liu J., Zhao L., Xue Y., Shi J., Suo L., Luo Y., Chai B., Yang C., Fang Q., Zhang Y., Bao Y., Pickens C. L., Lu L. 2014) **An unconditioned stimulus retrieval extinction procedure to prevent the**

return of fear memory *Biological psychiatry* **76**:895–901 https://doi.org/10.1016/j.biopsych.2014.03.027

MacLeod M. D., Macrae C. N. 2001) **Gone but not forgotten: the transient nature of retrieval-induced forgetting** *Psychological Science* **12**:148–152 https://doi.org/10.1111/1467-9280.00325

Meier B., König A., Parak S., Henke K. 2011) **Suppressed, but not forgotten** *Swiss Journal of Psychology / Schweizerische Zeitschrift für Psychologie / Revue Suisse de Psychologie* **70**:5–11 https://doi.org/10.1024/1421-0185/a000033

Merlo E., Milton A. L., Goozée Z. Y., Theobald D. E., Everitt B. J. 2014) **Reconsolidation and extinction are dissociable and mutually exclusive processes: behavioral and molecular evidence** *The Journal of neuroscience* **34**:2422–2431 https://doi.org/10.1523/jneurosci.4001-13.2014

Monfils M. H., Cowansage K. K., Klann E., LeDoux J. E. 2009) **Extinction-reconsolidation boundaries: key to persistent attenuation of fear memories** *Science* **324**:951–955 https://doi.org/10.1126/science.1167975

Nader K., Schafe G. E., Le Doux J. E. 2000) **Fear memories require protein synthesis in the amygdala for reconsolidation after retrieval** *Nature* **406**:722–726 https://doi.org/10.1038/35021052

Nader K., Schafe G. E., LeDoux J. E. 2000) **Fear memories require protein synthesis in the amygdala for reconsolidation after retrieval** *Nature* **406**:722–726 https://doi.org/10.1038/35021052

Penolazzi B., Stramaccia D. F., Braga M., Mondini S., Galfano G. 2014) **Human memory retrieval and inhibitory control in the brain: beyond correlational evidence** *The Journal of neuroscience* **34**:6606–6610 https://doi.org/10.1523/jneurosci.0349-14.2014

Raio C. M., Hartley C. A., Orederu T. A., Li J., Phelps E. A. 2017) **Stress attenuates the flexible updating of aversive value** *Proceedings of the National Academy of Sciences of the United States of America* **114**:11241–11246 https://doi.org/10.1073/pnas.1702565114

Rossi S., Hallett M., Rossini P. M., Pascual-Leone A. 2009) **Safety, ethical considerations, and application guidelines for the use of transcranial magnetic stimulation in clinical practice and research** *Clin Neurophysiol* **120**:2008–2039 https://doi.org/10.1016/j.clinph.2009.08.016

Rossini P. M., Burke D., Chen R., Cohen L. G., Daskalakis Z., Di Iorio R., Di Lazzaro V., Ferreri F., Fitzgerald P. B., George M. S., Hallett M., Lefaucheur J. P., Langguth B., Matsumoto H., Miniussi C., Nitsche M. A., Pascual-Leone A., Paulus W., Rossi S., ..., Ziemann U. 2015) **Non-invasive electrical and magnetic stimulation of the brain, spinal cord, roots and peripheral nerves: Basic principles and procedures for routine clinical and research application** *An updated report from an I.F.C.N. Committee. Clin Neurophysiol* **126**:1071–1107 https://doi.org/10.1016/j.clinph.2015.02.001

Ryan T. J., Roy D. S., Pignatelli M., Arons A., Tonegawa S. 2015) **Memory. Engram cells retain memory under retrograde amnesia** *Science* **348**:1007–1013 https://doi.org/10.1126/science.aaa5542

Saunders J., MacLeod M. D. 2002) **New evidence on the suggestibility of memory: the role of retrieval-induced forgetting in misinformation effects** *J Exp Psychol Appl* **8**:127–142

Schacter D. L., Addis D. R., Hassabis D., Martin V. C., Spreng R. N., Szpunar K. K. 2012) **The future of memory: remembering, imagining, and the brain** *Neuron* **76**:677–694 https://doi.org/10.1016/j.neuron.2012.11.001

Schacter D. L., Koutstaal W., Norman K. A. 1996) **Can cognitive neuroscience illuminate the nature of traumatic childhood memories?** *Curr Opin Neurobiol* **6**:207–214 https://doi.org/10.1016/s0959-4388(96)80074-2

Schiller D., Kanen J. W., LeDoux J. E., Monfils M. H., Phelps E. A. 2013) **Extinction during reconsolidation of threat memory diminishes prefrontal cortex involvement** *Proceedings of the National Academy of Sciences of the United States of America* **110**:20040–20045 https://doi.org/10.1073/pnas.1320322110

Schiller D., LeDoux J. E., Phelps E. A. 2020) **Reply to Beckers, McIntosh and Chambers on the verification of 'preventing the return of fear using retrieval-extinction in humans'** *PsyArxiv* https://doi.org/10.31234/osf.io/jn6uw

Schiller D., Monfils M. H., Raio C. M., Johnson D. C., Ledoux J. E., Phelps E. A. 2010) **Preventing the return of fear in humans using reconsolidation update mechanisms** *Nature* **463**:49–53 https://doi.org/10.1038/nature08637

Schiller D., Raio C. M., Phelps E. A. 2012) **Extinction training during the reconsolidation window prevents recovery of fear** *J Vis Exp: e* **3893** https://doi.org/10.3791/3893

Schroyens N., Beckers T., Luyten L. 2023) **Appraising reconsolidation theory and its empirical validation** *Psychonomic Bulletin & Review* **30**:450–463 https://doi.org/10.3758/s13423-022-02173-2

Sevenster D., Beckers T., Kindt M. 2012) **Retrieval per se is not sufficient to trigger reconsolidation of human fear memory** *Neurobiol Learn Mem* **97**:338–345 https://doi.org/10.1016/j.nlm.2012.01.009

Sevenster D., Beckers T., Kindt M. 2013) **Prediction error governs pharmacologically induced amnesia for learned fear** *Science* **339**:830–833 https://doi.org/10.1126/science.1231357

Speer M. E., Ibrahim S., Schiller D., Delgado M. R. 2021) **Finding positive meaning in memories of negative events adaptively updates memory** *Nature communications* **12**:6601 https://doi.org/10.1038/s41467-021-26906-4

Storm B., Bjork E., Bjork R. 2012) **On the durability of retrieval-induced forgetting** *Journal of Cognitive Psychology* **24**:617–629 https://doi.org/10.1080/20445911.2012.674030

Stramaccia D. F., Meyer A. K., Rischer K. M., Fawcett J. M., Benoit R. G. 2021) **Memory suppression and its deficiency in psychological disorders: A focused meta-analysis** *Journal of experimental psychology. General* **150**:828–850 https://doi.org/10.1037/xge0000971

Stramaccia D. F., Penolazzi B., Altoè G., Galfano G. 2017) **TDCS over the right inferior frontal gyrus disrupts control of interference in memory: A retrieval-induced forgetting study** *Neurobiol Learn Mem* **144**:114–130 https://doi.org/10.1016/j.nlm.2017.07.005

Su S., Deng J., Yuan K., Gong Y., Zhang Y., Li H., Cao K., Huang X., Lin X., Wu P., Xue Y., Bao Y., Shi J., Shi L., Lu L. 2022) **Continuous theta-burst stimulation over the right dorsolateral prefrontal cortex disrupts fear memory reconsolidation in humans** *iScience* **25**:103614 https://doi.org/10.1016/j.isci.2021.103614

Tonegawa S., Liu X., Ramirez S., Redondo R. 2015) **Memory Engram Cells Have Come of Age** *Neuron* **87**:918–931 https://doi.org/10.1016/j.neuron.2015.08.002

Tonegawa S., Pignatelli M., Roy D. S., Ryan T. J. 2015) **Memory engram storage and retrieval** *Curr Opin Neurobiol* **35**:101–109 https://doi.org/10.1016/j.conb.2015.07.009

Valle T. M., Bajo M. T., Gómez-Ariza C. J. 2020) **Cathodal transcranial direct current stimulation over the right dorsolateral prefrontal cortex cancels out the cost of selective retrieval on subsequent analogical reasoning** *Neuropsychologia* **141**:107431 https://doi.org/10.1016/j.neuropsychologia.2020.107431

Wang Y., Zhu Z., Hu J., Schiller D., Li J. 2021) **Active suppression prevents the return of threat memory in humans** *Commun Biol* **4**:609 https://doi.org/10.1038/s42003-021-02120-2

Wells A., Davies M. I. 1994) **The Thought Control Questionnaire: a measure of individual differences in the control of unwanted thoughts** *Behav Res Ther* **32**:871–878 https://doi.org/10.1016/0005-7967(94)90168-6

Wimber M., Alink A., Charest I., Kriegeskorte N., Anderson M. C. 2015) **Retrieval induces adaptive forgetting of competing memories via cortical pattern suppression** *Nature neuroscience* **18**:582–589 https://doi.org/10.1038/nn.3973

Author information

Yinmei Ni*

School of Psychological and Cognitive Sciences and Beijing Key Laboratory of Behavior and Mental Health, Peking University, Beijing, China

ORCID iD: 0000-0003-3614-1828

*Equal contribution

Ye Wang*

School of Psychological and Cognitive Sciences and Beijing Key Laboratory of Behavior and Mental Health, Peking University, Beijing, China

*Equal contribution

Zijian Zhu

School of Psychology, Shaanxi Normal University, Xi'an, China

Jingchu Hu

Department of Anxiety Disorders, Shenzhen Kangning Hospital, Shenzhen Mental Health Institute, Shenzhen, China

Daniela Schiller

Departments of Psychiatry and Neuroscience, and Friedman Brain Institute, Icahn School of Medicine at Mount Sinai, New York, United States

ORCID iD: 0000-0002-0357-7724

Jian Li
School of Psychological and Cognitive Sciences and Beijing Key Laboratory of Behavior and Mental Health, Peking University, Beijing, China, IDG/McGovern Institute for Brain Research, Peking University, Beijing, China
ORCID iD: 0000-0002-3941-2622

For correspondence: leekin@gmail.com

Editors

Reviewing Editor
Mihaela Iordanova
Concordia University, Montreal, Canada

Senior Editor
Yanchao Bi
Beijing Normal University, Beijing, China

Reviewer #1 (Public review):

Summary:

The novel advance by Wang et al is in the demonstration that, relative to a standard extinction procedure, the retrieval-extinction procedure more effectively suppresses responses to a conditioned threat stimulus when testing occurs just minutes after extinction. The authors provide some solid evidence to show that this "short-term" suppression of responding involves engagement of the dorsolateral prefrontal cortex.

Strengths:

Overall, the study is well-designed and the results are potentially interesting. There are, however, a few issues in the way that it is introduced and discussed. Some of the issues concern clarity of expression/communication. However, others relate to a theory that could be used to help the reader understand why the results should have come out the way that they did. More specific comments and questions are presented below.

Weaknesses:

INTRODUCTION & THEORY

(1) It is difficult to appreciate why the first trial of extinction in a standard protocol does NOT produce the retrieval-extinction effect. This applies to the present study as well as others that have purported to show a retrieval-extinction effect. The importance of this point comes through at several places in the paper. E.g., the two groups in Study 1 experienced a different interval between the first and second CS extinction trials; and the results varied with this interval: a longer interval (10 min) ultimately resulted in less reinstatement of fear than a shorter interval. Even if the different pattern of results in these two groups was shown/known to imply two different processes, there is nothing in the present study that addresses what those processes might be. That is, while the authors talk about mechanisms of memory updating, there is little in the present study that permits any clear statement about mechanisms of memory. The references to a "short-term memory update" process do not help the reader to understand what is happening in the protocol.

In reply to this point, the authors cite evidence to suggest that "an isolated presentation of the CS+ seems to be important in preventing the return of fear expression." They then note the following: "It has also been suggested that only when the old memory and new experience (through extinction) can be inferred to have been generated from the same underlying latent cause, the old memory can be successfully modified(Gershman et al., 2017). On the other hand, if the new experiences are believed to be generated by a different latent cause, then the old memory is less likely to be subject to modification. Therefore, the way the 1stand 2ndCS are temporally organized (retrieval-extinction or standard extinction) might affect how the latent cause is inferred and lead to different levels of fear expression from a theoretical perspective." This merely begs the question: why might an isolated presentation of the CS+ result in the subsequent extinction experiences being allocated to the same memory state as the initial conditioning experiences? This is not yet addressed in any way.

(2) The discussion of memory suppression is potentially interesting but, in its present form, raises more questions than it answers. That is, memory suppression is invoked to explain a particular pattern of results but I, as the reader, have no sense of why a fear memory would be better suppressed shortly after the retrieval-extinction protocol compared to the standard extinction protocol; and why this suppression is NOT specific to the cue that had been subjected to the retrieval-extinction protocol.

(3) Relatedly, how does the retrieval-induced forgetting (which is referred to at various points throughout the paper) relate to the retrieval-extinction effect? The appeal to retrieval-induced forgetting as an apparent justification for aspects of the present study reinforces points 2 and 3 above. It is not uninteresting but lacks clarification/elaboration and, therefore, its relevance appears superficial at best.

(4) I am glad that the authors have acknowledged the papers by Chalkia, van Oudenhove & Beckers (2020) and Chalkia et al (2020), which failed to replicate the effects of retrieval-extinction reported by Schiller et al in Reference 6. The authors have inserted the following text in the revised manuscript: "It should be noted that while our long-term amnesia results were consistent with the fear memory reconsolidation literature, there were also studies that failed to observe fear prevention (Chalkia, Schroyens, et al., 2020; Chalkia, Van Oudenhove, et al., 2020; Schroyens et al., 2023). Although the memory reconsolidation framework provides a viable explanation for the long-term amnesia, more evidence is required to validate the presence of reconsolidation, especially at the neurobiological level (Elsey et al., 2018). While it is beyond the scope of the current study to discuss the discrepancies between these studies, one possibility to reconcile these results concerns the procedure for the retrieval-extinction training. It has been shown that the eligibility for old memory to be updated is contingent on whether the old memory and new observations can be inferred to have been generated by the same latent cause (Gershman et al., 2017; Gershman and Niv, 2012). For example, prevention of the return of fear memory can be achieved through gradual extinction paradigm, which is thought to reduce the size of prediction errors to inhibit the formation of new latent causes (Gershman, Jones, et al., 2013). Therefore, the effectiveness of the retrieval-extinction paradigm might depend on the reliability of such paradigm in inferring the same underlying latent cause." Firstly, if it is beyond the scope of the present study to discuss the discrepancies between the present and past results, it is surely beyond the scope of the study to make any sort of reference to clinical implications!!! Secondly, it is perfectly fine to state that "the effectiveness of the retrieval-extinction paradigm might depend on the reliability of such paradigm in inferring the same underlying latent cause..." This is not uninteresting, but it also isn't saying much. Minimally, I would expect some statement about factors that are likely to determine whether one is or isn't likely to see a retrieval-extinction effect, grounded in terms of this theory.

CLARIFICATIONS, ELABORATIONS, EDITS

(5) Some parts of the paper are not easy to follow. Here are a few examples (though there are others):

(a) In the abstract, the authors ask "whether memory retrieval facilitates update mechanisms other than memory reconsolidation"... but it is never made clear how memory retrieval could or should "facilitate" a memory update mechanism.

(b) The authors state the following: "Furthermore, memory reactivation also triggers fear memory reconsolidation and produces cue specific amnesia at a longer and separable timescale (Study 2, N = 79 adults)." Importantly, in study 2, the retrieval-extinction protocol produced a cue-specific disruption in responding when testing occurred 24 hours after the end of extinction. This result is interesting but cannot be easily inferred from the statement that begins "Furthermore..." That is, the results should be described in terms of the combined effects of retrieval and extinction, not in terms of memory reactivation alone; and the statement about memory reconsolidation is unnecessary. One can simply state that the retrieval-extinction protocol produced a cue-specific disruption in responding when testing occurred 24 hours after the end of extinction.

(c) The authors also state that: "The temporal scale and cue-specificity results of the short-term fear amnesia are clearly dissociable from the amnesia related to memory reconsolidation, and suggest that memory retrieval and extinction training trigger distinct underlying memory update mechanisms." ***The pattern of results when testing occurred just minutes after the retrieval-extinction protocol was different to that obtained when testing occurred 24 hours after the protocol. Describing this in terms of temporal scale is unnecessary; and suggesting that memory retrieval and extinction trigger different memory update mechanisms is not obviously warranted. The results of interest are due to the combined effects of retrieval+extinction and there is no sense in which different memory update mechanisms should be identified with the different pattern of results obtained when testing occurred either 30 min or 24 hours after the retrieval-extinction protocol (at least, not the specific pattern of results obtained here).

(d) The authors state that: "We hypothesize that the labile state triggered by the memory retrieval may facilitate different memory update mechanisms following extinction training, and these mechanisms can be further disentangled through the lens of temporal dynamics and cue-specificities." *** The first part of the sentence is confusing around usage of the term "facilitate"; and the second part of the sentence that references a "lens of temporal dynamics and cue-specificities" is mysterious. Indeed, as all rats received the same retrieval-extinction exposures in Study 2, it is not clear how or why any differences between the groups are attributed to "different memory update mechanisms following extinction".

DATA

(6A) The eight participants who were discontinued after Day 1 in Study 1 were all from the no reminder group. The authors should clarify how participants were allocated to the two groups in this experiment so that the reader can better understand why the distribution of non-responders was non-random (as it appears to be).

(6B) Similarly, in study 2, of the 37 participants that were discontinued after Day 2, 19 were from Group 30 min and 5 were from Group 6 hours. The authors should comment on how likely these numbers are to have been by chance alone. I presume that they reflect something about the way that participants were allocated to groups: e.g., the different groups of participants in studies 1 and 2 could have been run at quite different times (as opposed to concurrently). If this was done, why was it done? I can't see why the study should have been conducted in this fashion - this is for myriad reasons, including the authors' concerns re SCRs and their seasonal variations.

(6C) In study 2, why is responding to the CS- so high on the first test trial in Group 30 min? Is the change in responding to the CS- from the last extinction trial to the first test trial different across the three groups in this study? Inspection of the figure suggests that it is higher in

Group 30 min relative to Groups 6 hours and 24 hours. If this is confirmed by the analysis, it has implications for the fear recovery index which is partly based on responses to the CS-. If not for differences in the CS- responses, Groups 30 min and 6 hours are otherwise identical. That is, the claim of differential recovery to the CS1 and CS2 across time may simply an artefact of the way that the recovery index was calculated. This is unfortunate but also an important feature of the data given the way in which the fear recovery index was calculated.

(6D) The 6 hour group was clearly tested at a different time of day compared to the 30 min and 24 hour groups. This could have influenced the SCRs in this group and, thereby, contributed to the pattern of results obtained.

(6E) The authors find different patterns of responses to CS1 and CS2 when they were tested 30 min after extinction versus 24 h after extinction. On this basis, they infer distinct memory update mechanisms. However, I still can't quite see why the different patterns of responses at these two time points after extinction need to be taken to infer different memory update mechanisms. That is, the different patterns of responses at the two time points could be indicative of the same "memory update mechanism" in the sense that the retrieval-extinction procedure induces a short-term memory suppression that serves as the basis for the longer-term memory suppression (i.e., the reconsolidation effect). My pushback on this point is based on the notion of what constitutes a memory update mechanism; and is motivated by what I take to be a rather loose use of language/terminology in the reconsolidation literature and this paper specifically (for examples, see the title of the paper and line 2 of the abstract).

https://doi.org/10.7554/eLife.98652.2.sa2

Reviewer #2 (Public review):

Summary

The study investigated whether memory retrieval followed soon by extinction training results in a short-term memory deficit when tested - with a reinstatement test that results in recovery from extinction - soon after extinction training. Experiment 1 documents this phenomenon using a between-subjects design. Experiment 2 used a within-subject control and saw that the effect is also observed in a control condition. In addition, it also revealed that if testing is conducted 6 hours after extinction, there is not effect of retrieval prior to extinction as there is recovery from extinction independently of retrieval prior to extinction. A third Group also revealed that retrieval followed by extinction attenuates reinstatement when the test is conducted 24 hours later, consistent with previous literature. Finally, Experiment 3 used continuous theta-burst stimulation of the dorsolateral prefrontal cortex and assessed whether inhibition of that region (vs a control region) reversed the short-term effect revealed in Experiments 1 and 2. The results of control groups in Experiment 3 replicated the previous findings (short-term effect), and the experimental group revealed that these can be reversed by inhibition of the dorsolateral prefrontal cortex.

Strengths

The work is performed using standard procedures (fear conditioning and continuous theta-burst stimulation) and there is some justification of the sample sizes. The results replicate previous findings - some of which have been difficult to replicate and this needs to be acknowledged - and suggest that the effect can also be observed in a short-term reinstatement test.

The study establishes links between the memory reconsolidation and retrieval-induced forgetting (or memory suppression) literatures. The explanations that have been developed for these are distinct and the current results integrate these, by revealing that the DLPFC activity involved in retrieval-extinction short-term effect. There is thus some novelty in the present results, but numerous questions remain unaddressed.

Weakness

The fear acquisition data is converted to a differential fear SCR and this is what is analysed (early vs late). However, the figure shows the raw SCR values for CS+ and CS- and therefore it is unclear whether acquisition was successful (despite there being an "early" vs "late" effect - no descriptives are provided).

In Experiment 1 (Test results) it is unclear whether the main conclusion stems from a comparison of the test data relative to the last extinction trial ("we defined the fear recovery index as the SCR difference between the first test trial and the last extinction trial for a specific CS") or the difference relative to the CS- ("differential fear recovery index between CS+ and CS-"). It would help the reader assess the data if Fig 1e presents all the indexes (both CS+ and CS-). In addition, there is one sentence which I could not understand "there is no statistical difference between the differential fear recovery indexes between CS+ in the reminder and no reminder groups (P=0.048)". The p value suggests that there is a difference, yet it is not clear what is being compared here. Critically, any index taken as a difference relative to the CS- can indicate recovery of fear to the CS+ or absence of discrimination relative to the CS-, so ideally the authors would want to directly compare responses to the CS+ in the reminder and no-reminder groups. In the absence of such comparison, little can be concluded, in particular if SCR CS- data is different between groups. The latter issue is particularly relevant in Experiment 2, in which the CS- seems to vary between groups during the test and this can obscure the interpretation of the result.

In experiment 1, the findings suggest that there is a benefit of retrieval followed by extinction in a short-term reinstatement test. In Experiment 2, the same effect is observed to a cue which did not undergo retrieval before extinction (CS2+), a result that is interpreted as resulting from cue-independence, rather than a failure to replicate in a within-subjects design the observations of Experiment 1 (between-subjects). Although retrieval-induced forgetting is cue-independent (the effect on items that are suppressed [Rp-] can be observed with an independent probe), it is not clear that the current findings are similar, and thus that the strong parallels made are not warranted. Here, both cues have been extinguished and therefore been equally exposed during the critical stage.

The findings in Experiment 2 suggest that the amnesia reported in Experiment 1 is transient, in that no effect is observed when the test is delayed by 6 hours. The phenomena whereby reactivated memories transition to extinguished memories as a function of the amount of exposure (or number of trials) is completely different from the phenomena observed here. In the former, the manipulation has to do with the number of trials (or total amount of time) that the cues are exposed. In the current Experiment 2, the authors did not manipulate the number of trials but instead the retention interval between extinction and test. The finding reported here is closer to a "Kamin effect", that is the forgetting of learned information which is observed with intervals of intermediate length (Baum, 1968). Because the Kamin effect has been inferred to result from retrieval failure, it is unclear how this can be explained here. There needs to be much more clarity on the explanations to substantiate the conclusions. There are many results (Ryan et al., 2015) that challenge the framework that the authors base their predictions on (consolidation and reconsolidation theory), therefore these need to be acknowledged. These studies showed that memory can be expressed in the absence of the biological machinery thought to be needed for memory performance. The authors should be careful about statements such as "eliminate fear memores" for which there is little evidence.

The parallels between the current findings and the memory suppression literature are speculated in the general discussion, and there is the conclusion that "the retrieval-extinction procedure might facilitate a spontaneous memory suppression process". Because one of the basic tenets of the memory suppression literature is that it reflects an "active suppression" process, there is no reason to believe that in the current paradigm the same phenomenon is in place, but instead it is "automatic". In other words, the conclusions make strong parallels with the memory suppression (and cognitive control) literature, yet the phenomena that they observed is thought to be passive (or spontaneous/automatic). Ultimately, it is unclear why 10 mins between the reminder and extinction learning will "automatically" suppress fear memories. Further down in the discussion it is argued that "For example, in the well-known retrieval-induced forgetting (RIF) phenomenon, the recall of a stored memory can impair the retention of related long-term memory and this forgetting effect emerges as early as 20 minutes after the retrieval procedure, suggesting memory suppression or inhibition can occur in a more spontaneous and automatic manner". I did not follow with the time delay between manipulation and test (20 mins) would speak about whether the process is controlled or automatic. In addition, the links with the "latent cause" theoretical framework are weak if any. There is little reason to believe that one extinction trial, separated by 10 mins from the rest of extinction trials, may lead participants to learn that extinction and acquisition have been generated by the same latent cause.

Among the many conclusions, one is that the current study uncovers the "mechanism" underlying the short-term effects of retrieval-extinction. There is little in the current report that uncovers the mechanism, even in the most psychological sense of the mechanism, so this needs to be clarified. The same applies to the use of "adaptive".

Whilst I could access the data in the OFS site, I could not make sense of the Matlab files as there is no signposting indicating what data is being shown in the files. Thus, as it stands, there is no way of independently replicating the analyses reported.

The supplemental material shows figures with all participants, but only some statistical analyses are provided, and sometimes these are different from those reported in the main manuscript. For example, the test data in Experiment 1 is analysed with a two-way ANOVA with main effects of group (reminder vs no-reminder) and time (last trial of extinction vs first trial of test) in the main report. The analyses with all participants in the sup mat used a mixed two-way ANOVA with group (reminder vs no reminder) and CS (CS+ vs CS-). This makes it difficult to assess the robustness of the results when including all participants. In addition, in the supplementary materials there are no figures and analyses for Experiment 3.

One of the overarching conclusions is that the "mechanisms" underlying reconsolidation (long term) and memory suppression (short term) phenomena are distinct, but memory suppression phenomena can also be observed after a 7-day retention interval (Storm et al., 2012), which then questions the conclusions achieved by the current study.

References:

Baum, M. (1968). Reversal learning of an avoidance response and the Kamin effect. Journal of Comparative and Physiological Psychology, 66(2), 495.
Chalkia, A., Schroyens, N., Leng, L., Vanhasbroeck, N., Zenses, A. K., Van Oudenhove, L., & Beckers, T. (2020). No persistent attenuation of fear memories in humans: A registered replication of the reactivation-extinction effect. Cortex, 129, 496-509.
Ryan, T. J., Roy, D. S., Pignatelli, M., Arons, A., & Tonegawa, S. (2015). Engram cells retain memory under retrograde amnesia. Science, 348(6238), 1007-1013.
Storm, B. C., Bjork, E. L., & Bjork, R. A. (2012). On the durability of retrieval-induced forgetting. Journal of Cognitive Psychology, 24(5), 617-629.

Comments on revisions:

The authors have revised the manuscript but most of my concerns have remained unaddressed.

(1) There are still no descriptive statistics to substantiate learning in Experiment 1.

(2) In the revised analyses, the authors now show that CS- changes in different groups (for example, Experiment 2) so this means that there is little to conclude from the differential scores because these depend on CS-. It is unclear whether the effects arise from CS+ performance or the differential which is subject to CS- variations.

(3) The notion that suppression is automatic is speculative at best

(4) It still struggle with the parallels between these findings and the "limbo" literature. Here you manipulated the retention interval, whereas in the cited studies the number of extinction (exposure) was varied. These are two completely different phenomena.

(5) My point about the data problematic for the reconsolidation (and consolidation) frameworks is that they observed memory in the absence of the brain substrates that are needed for memory to be observed. The answer did not address this. I do not understand how the latent cause model can explain this, if the only difference is the first ITI. Wouldn't participants fail to integrate extinction with acquisition with a longer ITI?

(6) The materials in the OSF site are the same as before, they haven't ben updated.

(7) Concerning supplementary materials, the robustness tests are intended to prove that you 1) can get the same results by varying the statistical models or 2) you can get the same results when you include all participants. Here authors have done both so this does not help. Also, in the rebuttal letter, they stated "Please note we did not include non-learners in these analyses " which contradicts what is stated in the figure captions "(learners + non learners)"

(8) Finally, the literature suggesting that reconsolidation interference "eliminates" a memory is not substantiated by data nor in line with current theorising, so I invite a revision of these strong claims.

Overall, I conclude that the revised manuscript did not address my main concerns.

https://doi.org/10.7554/eLife.98652.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Reviewer #1 (Public Review):***
>
> *Weaknesses:*
>
> *INTRODUCTION & THEORY*
>
> *(1) Can the authors please clarify why the first trial of extinction in a standard protocol does NOT produce the retrieval-extinction effect? Particularly as the results section states: "Importantly, such a short-term effect is also retrieval dependent, suggesting the labile state of memory is necessary for the short-term memory update to take effect (Fig. 1e)." The importance of this point comes through at several places in the paper:*
>
> *1A. "In the current study, fear recovery was tested 30 minutes after extinction training, whereas the effect of memory reconsolidation was generally evident only several hours later and possibly with the help of sleep, leaving open the possibility of a different cognitive mechanism for the short-term fear dementia related to the retrieval-extinction procedure." ***What does this mean? The two groups in study 1 experienced a different*

> *interval between the first and second CS extinction trials; and the results varied with this interval: a longer interval (10 min) ultimately resulted in less reinstatement of fear than a shorter interval. Even if the different pattern of results in these two groups was shown/known to imply two different processes, there is absolutely no reason to reference any sort of cognitive mechanism or dementia - that is quite far removed from the details of the present study.*

Indeed, the only difference between the standard extinction paradigm and the retrieval-extinction paradigm is the difference between the first and second CS extinction trials. It has been shown before that a second CS+ presented 1 hour after the initial retrieval CS+ resulted in the dephosphorylation of GluR1 in rats, which was indicative of memory destabilization. The second CS+ presented only 3 minutes after the initial retrieval CS+, as in the standard extinction training, did not cause the GluR1 dephosphorylation effect (Monfils et al., 2009). Therefore, an isolated presentation of the CS+ seems to be important in preventing the return of fear expression. Behaviorally, when the CSs were presented in a more temporally spaced (vs. mass presentation) or a more gradual manner in the extinction training, the fear amnesia effects were more salient (Cain et al., 2003, Gershman et al., 2013). It has also been suggested that only when the old memory and new experience (through extinction) can be inferred to have been generated from the same underlying latent cause, the old memory can be successfully modified (Gershman et al., 2017). On the other hand, if the new experiences are believed to be generated by a different latent cause, then the old memory is less likely to be subject to modification. Therefore, the way the first and 2nd CS are temporally organized (retrieval-extinction or standard extinction) might affect how the latent cause is inferred and lead to different levels of fear expression from a theoretical perspective. These findings, together with studies in both fear and drug memories using the retrieval-extinction paradigm (Liu et al., 2014, Luo et al., 2015, Schiller et al., 2010, Xue et al., 2012), seem to suggest that the retrieval-extinction and the standard extinction procedures engage different cognitive and molecular mechanisms that lead to significant different behavioral outcomes.

In our study, we focus on the short-term and long-term amnesia effects of the retrieval-extinction procedure but also point out the critical role of retrieval in eliciting the short-term effect.

> *1B. "Importantly, such a short-term effect is also retrieval dependent, suggesting the labile state of memory is necessary for the short-term memory update to take effect (Fig. 1e)." ***As above, what is "the short-term memory update"? At this point in the text, it would be appropriate for the authors to discuss why the retrieval-extinction procedure produces less recovery than a standard extinction procedure as the two protocols only differ in the interval between the first and second extinction trials. References to a "short-term memory update" process do not help the reader to understand what is happening in the protocol.*

Sorry for the lack of clarity here. By short-term memory update we meant the short-term amnesia in fear expression.

> *(2) "Indeed, through a series of experiments, we identified a short-term fear amnesia effect following memory retrieval, in addition to the fear reconsolidation effect that appeared much later."*
>
> ****The only reason for supposing two effects is because of the differences in responding to the CS2, which was subjected to STANDARD extinction, in the short- and long-term tests. More needs to be said about how and why the performance of CS2 is affected in the short-term test and recovers in the long-term test. That is, if the loss of performance to CS1 and CS2 is going to be attributed to some type of memory updating process across the retrieval-extinction procedure, one needs to explain the selective recovery of*

> *performance to CS2 when the extinction-to-testing interval extends to 24 hours. Instead of explaining this recovery, the authors note that performance to CS1 remains low when the extinction-to-testing interval is 24 hours and invoke something to do with memory reconsolidation as an explanation for their results: that is, they imply (I think) that reconsolidation of the CS1-US memory is disrupted across the 24-hour interval between extinction and testing even though CS1 evokes negligible responding just minutes after extinction.*

In our results, we did not only focus on the fear expression related to CS2. In fact, we also demonstrated that the CS1 related fear expression diminished in the short-term memory test but re-appeared in the long-term memory after the CS1 retrieval-extinction training.

The "…recovery of performance to CS2 when the extinction-to-testing interval extends to 24 hours…" is a result that has been demonstrated in various previous studies (Kindt and Soeter, 2018, Kindt et al., 2009, Nader et al., 2000, Schiller et al., 2013, Schiller et al., 2010, Xue et al., 2012). That is, the reconsolidation framework stipulates that the pharmacological or behavioral intervention during the labile states of the reconsolidation window only modifies the fear memory linked to the reminded retrieval cue, but not for the non-reminded CS-US memory expression (but also see (Liu et al., 2014, Luo et al., 2015) for using the unconditioned stimulus as the reminder cue and the retrieval-extinction paradigm to prevent the return of fear memory associated with different CS). In fact, we hypothesized the temporal dynamics of CS1 and CS2 related fear expressions were due to the interplay between the short-term and long-term (reconsolidation) effects of the retrieval-extinction paradigm in the last figure (Fig. 6).

> *(3) The discussion of memory suppression is potentially interesting but, in its present form, raises more questions than it answers. That is, memory suppression is invoked to explain a particular pattern of results but I, as the reader, have no sense of why a fear memory would be better suppressed shortly after the retrieval-extinction protocol compared to the standard extinction protocol; and why this suppression is NOT specific to the cue that had been subjected to the retrieval-extinction protocol.*

We discussed memory suppression as one of the potential mechanisms to account for the three characteristics of the short-term amnesia effects: cue-independence, temporal dynamics (short-term) and thought-control-ability relevance. According to the memory suppression theory, the memory suppression effect is NOT specific to the cue and this effect was demonstrated via the independent cue test in a variety of studies (Anderson and Floresco, 2022, Anderson and Green, 2001, Gagnepain et al., 2014, Zhu et al., 2022). Therefore, we suggest in the discussion that it might be possible the CS1 retrieval cue prompted an automatic suppression mechanism and yielded the short-term fear amnesia consistent with various predictions from the memory suppression theory:

"In our experiments, subjects were not explicitly instructed to suppress their fear expression, yet the retrieval-extinction training significantly decreased short-term fear expression. These results are consistent with the short-term amnesia induced with the more explicit suppression intervention (Anderson et al., 1994; Kindt and Soeter, 2018; Speer et al., 2021; Wang et al., 2021; Wells and Davies, 1994). It is worth noting that although consciously repelling unwanted memory is a standard approach in memory suppression paradigm, it is possible that the engagement of the suppression mechanism can be unconscious. For example, in the retrieval-induced forgetting (RIF) paradigm, recall of a stored memory impairs the retention of related target memory and this forgetting effect emerges as early as 20 minutes after the retrieval procedure, suggesting memory suppression or inhibition can occur in a more spontaneous and automatic manner (Imai et al., 2014). Moreover, subjects with trauma histories exhibited more suppression-induced forgetting for both negative and neutral memories than those with little or no trauma (Hulbert and Anderson, 2018). Similarly, people with higher self-reported thought-control capabilities showed more severe cue-independent memory recall deficit, suggesting that suppression mechanism is associated with individual differences in spontaneous control abilities over intrusive thoughts (Küpper et al., 2014). It has also been suggested that similar automatic mechanisms might be involved in organic retrograde amnesia of traumatic childhood memories (Schacter et al., 2012; Schacter et al., 1996)."

> *3A. Relatedly, how does the retrieval-induced forgetting (which is referred to at various points throughout the paper) relate to the retrieval-extinction effect? The appeal to retrieval-induced forgetting as an apparent justification for aspects of the present study reinforces points 2 and 3 above. It is not uninteresting but needs some clarification/elaboration.*

We introduced the retrieval-induced forgetting (RIF) to make the point that RIF was believed to be related to the memory suppression mechanism and the RIF effect can appear relatively early, consistent with what we observed in the short-term amnesia effect. We have re-written the manuscript to make this point clearer:

"It is worth noting that although consciously repelling unwanted memory is a standard approach in memory suppression paradigm, it is possible that the engagement of the suppression mechanism can be unconscious. For example, in the retrieval-induced forgetting (RIF) paradigm, recall of a stored memory impairs the retention of related target memory and this forgetting effect emerges as early as 20 minutes after the retrieval procedure, suggesting memory suppression or inhibition can occur in a more spontaneous and automatic manner (Imai et al., 2014). Moreover, subjects with trauma histories exhibited more suppression-induced forgetting for both negative and neutral memories than those with little or no trauma (Hulbert and Anderson, 2018). Similarly, people with higher self-reported thought-control capabilities showed more severe cue-independent memory recall deficit, suggesting that suppression mechanism is associated with individual differences in spontaneous control abilities over intrusive thoughts (Küpper et al., 2014)."

> *(4) Given the reports by Chalkia, van Oudenhove & Beckers (2020) and Chalkia et al (2020), some qualification needs to be inserted in relation to reference 6. That is, reference 6 is used to support the statement that "during the reconsolidation window, old fear memory can be updated via extinction training following fear memory retrieval". This needs a qualifying statement like "[but see Chalkia et al (2020a and 2020b) for failures to reproduce the results of 6]."*
>
> *https://pubmed.ncbi.nlm.nih.gov/32580869/*
>
> *https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7115860/*

We have incorporated the reviewer's suggestion into the revised manuscript in both the introduction:

"Pharmacological blockade of protein synthesis and behavioral interventions can both eliminate the original fear memory expression in the long-term (24 hours later) memory test (Lee, 2008; Lee et al., 2017; Schiller et al., 2013; Schiller et al., 2010), resulting in the cue-specific fear memory deficit (Debiec et al., 2002; Lee, 2008; Nader, Schafe, & LeDoux, 2000). For example, during the reconsolidation window, retrieving a fear memory allows it to be updated through extinction training (i.e., the retrieval-extinction paradigm (Lee, 2008; Lee et al., 2017; Schiller et al., 2013; Schiller et al., 2010), but also see (Chalkia, Schroyens, et al., 2020; Chalkia, Van Oudenhove, et al., 2020; D. Schiller, LeDoux, & Phelps, 2020)"

And in the discussion:

“It should be noted that while our long-term amnesia results were consistent with the fear memory reconsolidation literatures, there were also studies that failed to observe fear prevention (Chalkia, Schroyens, et al., 2020; Chalkia, Van Oudenhove, et al., 2020; Schroyens et al., 2023). Although the memory reconsolidation framework provides a viable explanation for the long-term amnesia, more evidence is required to validate the presence of reconsolidation, especially at the neurobiological level (Elsey et al., 2018). While it is beyond the scope of the current study to discuss the discrepancies between these studies, one possibility to reconcile these results concerns the procedure for the retrieval-extinction training. It has been shown that the eligibility for old memory to be updated is contingent on whether the old memory and new observations can be inferred to have been generated by the same latent cause (Gershman et al., 2017; Gershman and Niv, 2012). For example, prevention of the return of fear memory can be achieved through gradual extinction paradigm, which is thought to reduce the size of prediction errors to inhibit the formation of new latent causes (Gershman, Jones, et al., 2013). Therefore, the effectiveness of the retrieval-extinction paradigm might depend on the reliability of such paradigm in inferring the same underlying latent cause. Furthermore, other studies highlighted the importance of memory storage per se and suggested that memory retention was encoded in the memory engram cell ensemble connectivity whereas the engram cell synaptic plasticity is crucial for memory retrieval (Ryan et al., 2015; Tonegawa, Liu, et al., 2015; Tonegawa, Pignatelli, et al., 2015). It remains to be tested how the cue-independent short-term and cue-dependent long-term amnesia effects we observed could correspond to the engram cell synaptic plasticity and functional connectivity among engram cell ensembles (Figure 6). This is particularly important, since the cue-independent characteristic of the short-term amnesia suggest that either different memory cues fail to evoke engram cell activities, or the retrieval-extinction training transiently inhibits connectivity among engram cell ensembles. Finally, SCR is only one aspect of the fear expression, how the retrieval-extinction paradigm might affect subjects’ other emotional (such as the startle response) and cognitive fear expressions such as reported fear expectancy needs to be tested in future studies since they do not always align with each other (Kindt et al., 2009; Sevenster et al., 2012, 2013).”

> *5A. What does it mean to ask: "whether memory retrieval facilitates update mechanisms other than memory reconsolidation"? That is, in what sense could or would memory retrieval be thought to facilitate a memory update mechanism?*

It is widely documented in the literatures that memory retrieval renders the old memory into a labile state susceptible for the memory reconsolidation process. However, as we mentioned in the manuscript, studies have shown that memory reconsolidation requires the *de novo* protein synthesis and usually takes hours to complete. What remains unknown is whether old memories are subject to modifications other than the reconsolidation process. Our task specifically tested the short-term effect of the retrieval-extinction paradigm and found that fear expression diminished 30mins after the retrieval-extinction training. Such an effect cannot be accounted for by the memory reconsolidation effect.

> *5B. "First, we demonstrate that memory reactivation prevents the return of fear shortly after extinction training in contrast to the memory reconsolidation effect which takes several hours to emerge and such a short-term amnesia effect is cue independent (Study 1, N = 57 adults)."*
>
> ****The phrasing here could be improved for clarity: "First, we demonstrate that the retrieval-extinction protocol prevents the return of fear shortly after extinction training (i.e., when testing occurs just min after the end of extinction)." Also, cue-dependence of the retrieval-extinction effect was assessed in study 2.*

We thank the reviewer and have modified the phrasing of the sentence:

"First, we demonstrate that memory retrieval-extinction protocol prevents the return of fear expression shortly after extinction training and this short-term effect is memory reactivation dependent (Study 1, *N* = 57 adults)."

> *5C. "Furthermore, memory reactivation also triggers fear memory reconsolidation and produces cue-specific amnesia at a longer and separable timescale (Study 2, N = 79 adults)." ***In study 2, the retrieval-extinction protocol produced a cue-specific disruption in responding when testing occurred 24 hours after the end of extinction. This result is interesting but cannot be easily inferred from the statement that begins "Furthermore..." That is, the results should be described in terms of the combined effects of retrieval and extinction, not in terms of memory reactivation alone; and the statement about memory reconsolidation is unnecessary. One can simply state that the retrieval-extinction protocol produced a cue-specific disruption in responding when testing occurred 24 hours after the end of extinction.*

We have revised the text according to the reviewer's comment.

"Furthermore, across different timescales, the memory retrieval-extinction paradigm triggers distinct types of fear amnesia in terms of cue-specificity and cognitive control dependence, suggesting that the short-term fear amnesia might be caused by different mechanisms from the cue-specific amnesia at a longer and separable timescale (Study 2, *N* = 79 adults)."

> *5D. "...we directly manipulated brain activities in the dorsolateral prefrontal cortex and found that both memory retrieval and intact prefrontal cortex functions were necessary for the short-term fear amnesia."*
>
> ****This could be edited to better describe what was shown: E.g., "...we directly manipulated brain activities in the dorsolateral prefrontal cortex and found that intact prefrontal cortex functions were necessary for the short-term fear amnesia after the retrieval-extinction protocol."*

Edited:

"Finally, using continuous theta-burst stimulation (Study 3, *N* = 75 adults), we directly manipulated brain activity in the dorsolateral prefrontal cortex, and found that both memory reactivation and intact prefrontal cortex function were necessary for the short-term fear amnesia after the retrieval-extinction protocol."

> *5E. "The temporal scale and cue-specificity results of the short-term fear amnesia are clearly dissociable from the amnesia related to memory reconsolidation, and suggest that memory retrieval and extinction training trigger distinct underlying memory update mechanisms."*
>
> ****The pattern of results when testing occurred just minutes after the retrieval-extinction protocol was different from that obtained when testing occurred 24 hours after the protocol. Describing this in terms of temporal scale is unnecessary, and suggesting that memory retrieval and extinction trigger different memory update mechanisms is not obviously warranted. The results of interest are due to the combined effects of retrieval+extinction and there is no sense in which different memory update mechanisms should be identified with retrieval (mechanism 1) and extinction (mechanism 2).*

We did not argue for different memory update mechanisms for the "retrieval (mechanism 1) and extinction (mechanism 2)" in our manuscript. Instead, we proposed that the retrieval-extinction procedure, which was mainly documented in the previous literatures for its association with the reconsolidation-related fear memory retention (the long-term effect), also had a much faster effect (the short-term effect). These two effects differed in many aspects, suggesting that different memory update mechanisms might be involved.

> *5F. "These findings raise the possibility of concerted memory modulation processes related to memory retrieval..."*
>
> ****What does this mean?*

As we mentioned in our response to the previous comment, we believe that the retrieval-extinction procedure triggers different types of memory update mechanisms working on different temporal scales.

> *(6) "...suggesting that the fear memory might be amenable to a more immediate effect, in addition to what the memory reconsolidation theory prescribes..."*
>
> ****What does it mean to say that the fear memory might be amenable to a more immediate effect?*

We intended to state that the retrieval-extinction procedure can produce a short-term amnesia effect and have thus revised the text.

> > *(7) "Parallel to the behavioral manifestation of long- and short-term memory deficits, concurrent neural evidence supporting memory reconsolidation theory emphasizes the long-term effect of memory retrieval by hypothesizing that synapse degradation and de novo protein synthesis are required for reconsolidation."*
> >
> > ****This sentence needs to be edited for clarity.*

We have rewritten this sentence:

"Corresponding to the long-term behavioral manifestation, concurrent neural evidence supporting memory reconsolidation hypothesis emphasizes that synapse degradation and *de novo* protein synthesis are required for reconsolidation."

> *(8) "previous behavioral manipulations engendering the short-term declarative memory effect..."*
>
> ****What is the declarative memory effect? It should be defined.*

We meant the amnesia on declarative memory research, such as the memory deficit caused by the think/no-think paradigms. Texts have been modified for clarity:

"On the contrary, previous behavioral manipulations engendering the short-term amnesia on declarative memory, such as the think/no-think paradigm, hinges on the intact activities in brain areas such as dorsolateral prefrontal cortex (cognitive control) and its functional coupling with specific brain regions such as hippocampus (memory retrieval) (Anderson and Green, 2001; Wimber et al., 2015)."

> *(9) "The declarative amnesia effect emerges much earlier due to the online functional activity modulation..."*
>
> ****Even if the declarative memory amnesia effect had been defined, the reference to online functional activity modulation is not clear.*

We have rephrased the sentence:

“The declarative amnesia effect arises much earlier due to the more instant modulation of functional connectivity, rather than the slower processes of new protein synthesis in these brain regions.”

> *(10) "However, it remains unclear whether memory retrieval might also precipitate a short-term amnesia effect for the fear memory, in addition to the long-term prevention orchestrated by memory consolidation."*
>
> ****I found this sentence difficult to understand on my first pass through the paper. I think it is because of the phrasing of memory retrieval. That is, memory retrieval does NOT precipitate any type of short-term amnesia for the fear memory: it is the retrieval-extinction protocol that produces something like short-term amnesia. Perhaps this sentence should also be edited for clarity.*

We have changed “memory retrieval” to “retrieval-extinction” where applicable.

> *I will also note that the usage of "short-term" at this point in the paper is quite confusing: Does the retrieval-extinction protocol produce a short-term amnesia effect, which would be evidenced by some recovery of responding to the CS when tested after a sufficiently long delay? I don't believe that this is the intended meaning of "short-term" as used throughout the majority of the paper, right?*

By “short-term”, we meant the lack of fear expression in the test phase (measured by skin conductance responses) shortly after the retrieval-extinction procedure (30 mins in studies 1 & 2 and 1 hour in study 3). It does not indicate that the effect is by itself “short-lived”.

> *(11) "To fully comprehend the temporal dynamics of the memory retrieval effect..." ***What memory retrieval effect? This needs some elaboration.*

We’ve changed the phrase “memory retrieval effect” to “retrieval-extinction effect” to refer to the effect of retrieval-extinction on fear amnesia.

> *(12) "We hypothesize that the labile state triggered by the memory retrieval may facilitate different memory update mechanisms following extinction training, and these mechanisms can be further disentangled through the lens of temporal dynamics and cue-specificities."*
>
> ****What does this mean? The first part of the sentence is confusing around the usage of the term "facilitate"; and the second part of the sentence that references a "lens of temporal dynamics and cue-specificities" is mysterious. Indeed, as all rats received the same retrieval-extinction exposures in Study 2, it is not clear how or why any differences between the groups are attributed to "different memory update mechanisms following extinction".*

As the reviewer mentioned, if only one time point data were collected, we cannot differentiate whether different memory update mechanisms are involved. In study 2, however, the 3 groups only differed on the time onsets the reinstatement test was conducted. Accordingly, our results showed that the fear amnesia effects for CS1 and CS2 cannot be simply explained by forgetting: different memory update mechanisms must be at work to explain the characteristics of the SCR related to both CS1 and CS2 at three different time scales (30min, 6h and 24h). It was based on these results, together with the results from the TMS study (study 3), that we proposed the involvement of a short-term memory update mechanism in addition to the reconsolidation related fear amnesia (which should become evident much later) induced by the retrieval-extinction protocol.

> *(13) "In the first study, we aimed to test whether there is a short-term amnesia effect of fear memory retrieval following the fear retrieval-extinction paradigm."*
>
> ****Again, the language is confusing. The phrase, "a short-term amnesia effect" implies that the amnesia itself is temporary; but I don't think that this implication is intended. The problem is specifically in the use of the phrase "a short-term amnesia effect of fear memory retrieval." To the extent that short-term amnesia is evident in the data, it is not due to retrieval per se but, rather, the retrieval-extinction protocol.*

We have changed the wordings and replaced “memory retrieval” with “retrieval-extinction” where applicable.

> (14) The authors repeatedly describe the case where there was a 24-hour interval between extinction and testing as consistent with previous research on fear memory reconsolidation. Which research exactly? That is, in studies where a CS re-exposure was combined with a drug injection, responding to the CS was disrupted in a final test of retrieval from long-term memory which typically occurred 24 hours after the treatment. Is that what the authors are referring to as consistent? If so, which aspect of the results are consistent with those previous findings? Perhaps the authors mean to say that, in the case where there was a 24-hour interval between extinction and testing, the results obtained here are consistent with previous research that has used the retrieval-extinction protocol. This would clarify the intended meaning greatly.

Our 24 hour test results after the retrieval-extinction protocol was consistent with both pharmacological and behavioral intervention studies in fear memory reconsolidation studies (Kindt and Soeter, 2018, Kindt et al., 2009, Liu et al., 2014, Luo et al., 2015, Monfils et al., 2009, Nader et al., 2000, Schiller et al., 2013, Schiller et al., 2010, Xue et al., 2012) since the final test phase typically occurred 24 hours after the treatment. At the 24-hour interval, the memory reconsolidation effect would become evident either via drug administration or behavioral intervention (extinction training).

> *DATA*
>
> *(15) Points about data:*
>
> *5A. The eight participants who were discontinued after Day 1 in study 1 were all from the no-reminder group. Can the authors please comment on how participants were allocated to the two groups in this experiment so that the reader can better understand why the distribution of non-responders was non-random (as it appears to be)?*
>
> *15B. Similarly, in study 2, of the 37 participants that were discontinued after Day 2, 19 were from Group 30 min, and 5 were from Group 6 hours. Can the authors comment on how likely these numbers are to have been by chance alone? I presume that they reflect something about the way that participants were allocated to groups, but I could be wrong.*

We went back and checked out data. As we mentioned in the supplementary materials, we categorized subjects as non-responders if their SCR response to any CS was less than 0.02 in Day 1 (fear acquisition). Most of the discontinued participants (non-responders) in the no-reminder group (study 1) and the 30min & 24 h groups (study 2) were when the heating seasons just ended or were yet to start, respectively. It has been documented that human body thermal conditions were related to the quality of the skin conductance response (SCR) measurements (Bauer et al., 2022, Vila, 2004). We suspect that the non-responders might be related to the body thermal conditions caused by the lack of central heating.

> *15C. "Post hoc t-tests showed that fear memories were resilient after regular extinction training, as demonstrated by the significant difference between fear recovery indexes of the CS+ and CS- for the no-reminder group (t26 = 7.441, P < 0.001; Fig. 1e), while subjects in the reminder group showed no difference of fear recovery between CS+ and CS- (t29 = 0.797, P = 0.432, Fig. 1e)."*
>
> ****Is the fear recovery index shown in Figure 1E based on the results of the first test trial only? How can there have been a "significant difference between fear recovery indexes of the CS+ and CS- for the no-reminder group" when the difference in responding to the CS+ and CS- is used to calculate the fear recovery index shown in 1E? What are the t-tests comparing exactly, and what correction is used to account for the fact that they are applied post-hoc?*

As we mentioned in the results section of the manuscript, the fear recovery index was defined as "the SCR difference between the first test trial and the last extinction trial of a specific CS". We then calculated the "differential fear recovery index" (figure legends of Fig. 1e) between CS+ and CS- for both the reminder and no-reminder groups. The post-hoc t-tests were used to examine whether there were significant fear recoveries (compare to 0) in both the reminder (t_{29} = 0.797, P = 0.432, Fig. 1e) and no-reminder (t_{26} = 7.441, P < 0.001; Fig. 1e) groups. We realize that the description of Bonferroni correction was not specified in the original manuscript and hence added in the revision where applicable.

> *15D. "Finally, there is no statistical difference between the differential fear recovery indexes between CS+ in the reminder and no reminder groups (t55 = -2.022, P = 0.048; Fig. 1c, also see Supplemental Material for direct test for the test phase)."*
>
> ****Is this statement correct - i.e., that there is no statistically significant difference in fear recovery to the CS+ in the reminder and no reminder groups? I'm sure that the authors would like to claim that there IS such a difference; but if such a difference is claimed, one would be concerned by the fact that it is coming through in an uncorrected t-test, which is the third one of its kind in this paragraph. What correction (for the Type 1 error rate) is used to account for the fact that the t-tests are applied post-hoc? And if no correction, why not?*

We are sorry about the typo. The reviewer was correct that we meant to claim here that "… there is a significant difference between the differential fear recovery indexes between CS+ in the reminder and no-reminder groups (t_{55} =- 2.022, P = 0.048; Fig. 1e)". Note that the t-test performed here was a confirmatory test following our two-way ANOVA with main effects of group (reminder vs. no-reminder) and time (last extinction trial vs. first test trial) on the differential CS SCR response (CS+ minus CS-) and we found a significant group x time interaction effect ($F_{1,55}$ = 4.087, P = 0.048, η^2 = 0.069). The significant difference between the differential fear recovery indexes was simply a re-plot of the interaction effect mentioned above and therefore no multiple correction is needed. We have reorganized the sequence of the sentences such that this t-test now directly follows the results of the ANOVA:

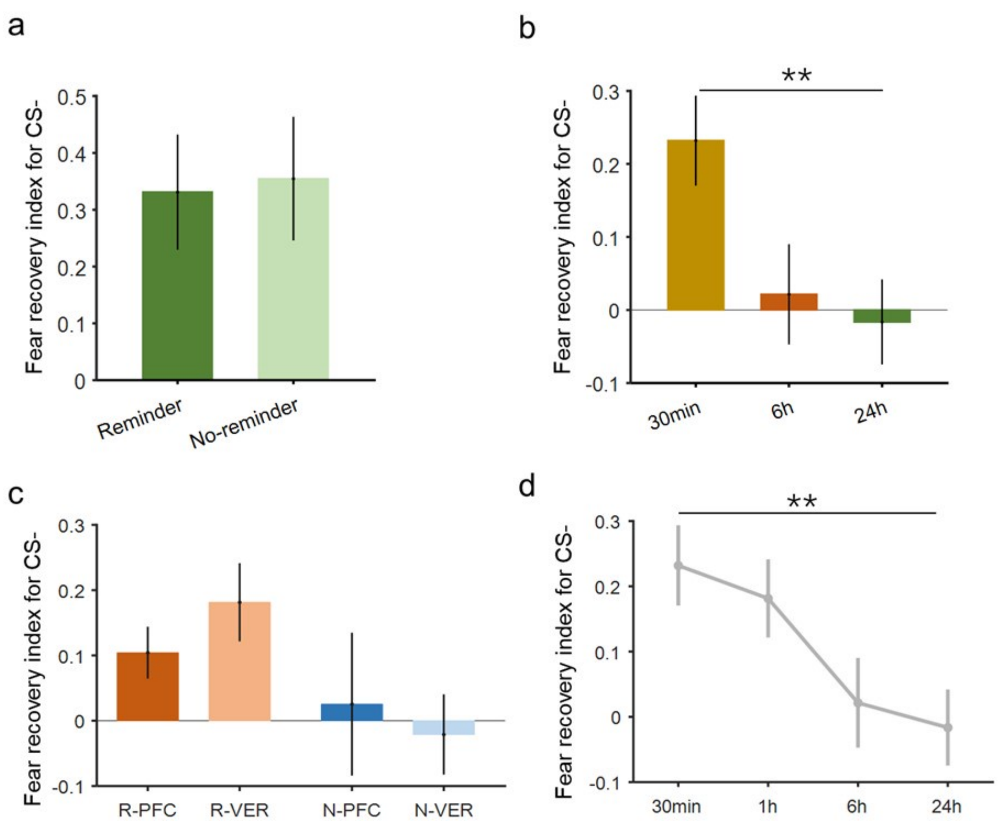
"The interaction effect was confirmed by the significant difference between the differential fear recovery indexes between CS1+ and CS2+ in the reminder and no-reminder groups (t_{55} = -2.022, P = 0.048; Figure 1E, also see Supplemental Material for the direct test of the test phase)."

> *15E. In study 2, why is responding to the CS- so high on the first test trial in Group 30 min? Is the change in responding to the CS- from the last extinction trial to the first test*

> *trial different across the three groups in this study? Inspection of the figure suggests that it is higher in Group 30 min relative to Groups 6 hours and 24 hours. If this is confirmed by the analysis, it has implications for the fear recovery index which is partly based on responses to the CS-. If not for differences in the CS- responses, Groups 30 minutes and 6 hours are otherwise identical.*

Following the reviewer's comments, we went back and calculated the mean SCR difference of CS- between the first test trial and the last extinction trial for all three studies (see Author response image 1 below). In study 1, there was no difference in the mean CS- SCR (between the first test trial and last extinction trial) between the reminder and no-reminder groups (Kruskal-Wallis test $\chi_1^2 = 0.230, P = 0.631$, panel a), though both groups showed significant fear recovery even in the CS- condition (Wilcoxon signed rank test, reminder: $P = 0.0043$, no-reminder: $P = 0.0037$). Next, we examined the mean SCR for CS- for the 30min, 6h and 24h groups in study 2 and found that there was indeed a group difference (one-way ANOVA,$F_{2.76} = 5.3462$, $P = 0.0067$, panel b), suggesting that the CS- related SCR was influenced by the test time (30min, 6h or 24h). We also tested the CS- related SCR for the 4 groups in study 3 (where test was conducted 1 hour after the retrieval-extinction training) and found that across TMS stimulation types (PFC vs. VER) and reminder types (reminder vs. no-reminder) the ANOVA analysis did not yield main effect of TMS stimulation type ($F_{1.71} = 0.322$, $P = 0.572$) nor main effect of reminder type ($F_{1.71} = 0.0499$, $P = 0.824$, panel c). We added the R-VER group results in study 3 (see panel c) to panel b and plotted the CS- SCR difference across 4 different test time points and found that CS- SCR decreased as the test-extinction delay increased (Jonckheere-Terpstra test, $P = 0.00028$). These results suggest a natural "forgetting" tendency for CS- related SCR and highlight the importance of having the CS- as a control condition to which the CS+ related SCR was compared with.

Author response image 1.



> *15F. Was the 6-hour group tested at a different time of day compared to the 30-minute and 24-hour groups; and could this have influenced the SCRs in this group?*

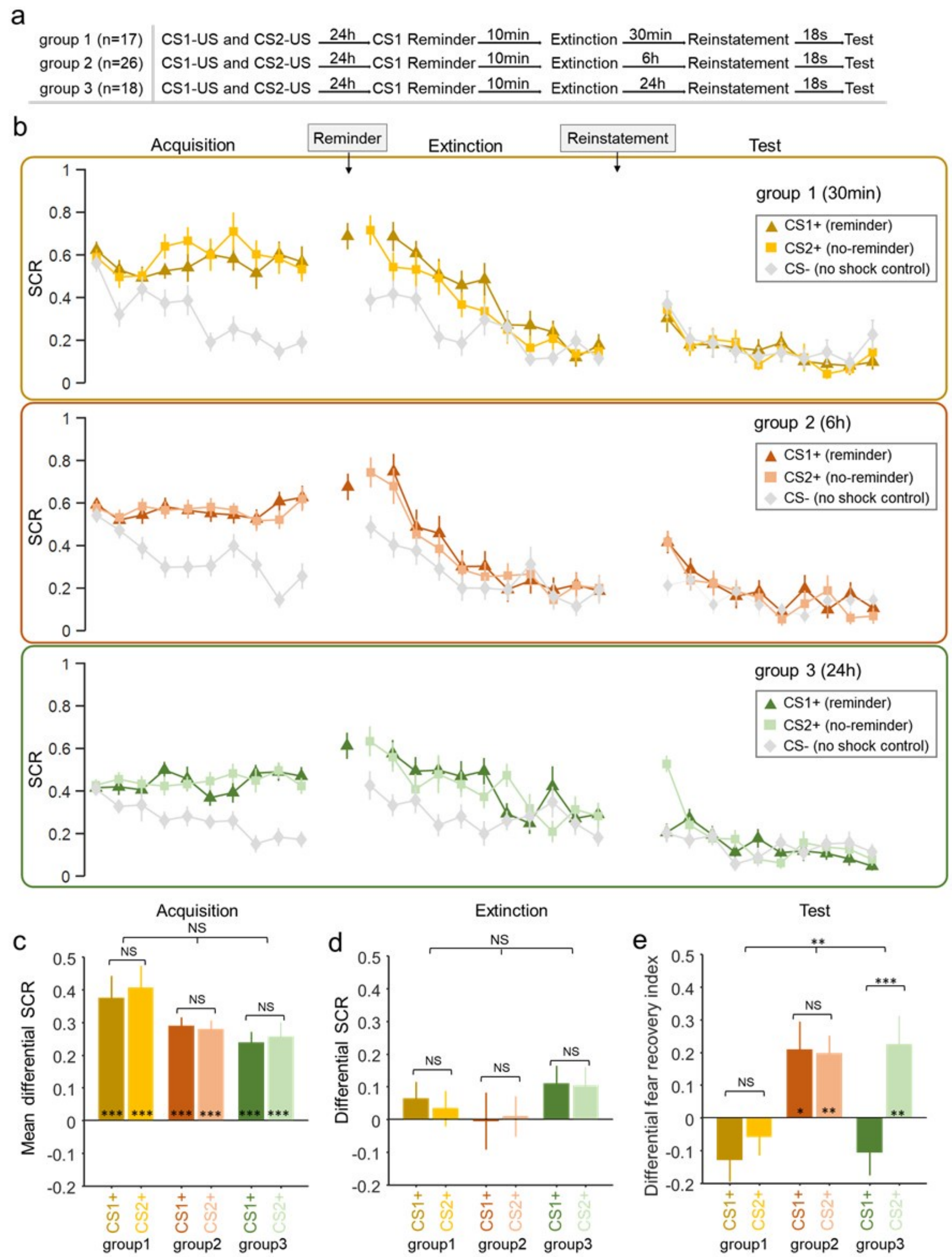
For the 30min and 24h groups, the test phase can be arranged in the morning, in the afternoon or at night. However, for the 6h group, the test phase was inevitably in the afternoon or at night since we wanted to exclude the potential influence of night sleep on the expression of fear memory (see Author response table 1 below). If we restricted the test time in the afternoon or at night for all three groups, then the timing of their extinction training was not matched.

Author response table 1.

Test time/Group	Morning	Afternoon	Night	Total# Subjects
30min	10	9	8	27
6h	0	12	14	26
24h	8	9	9	26

Nevertheless, we also went back and examined the data for the subjects only tested in the afternoon or at nights in the 30min and 24h groups to match with the 6h group where all the subjects were tested either in the afternoon or at night. According to Author response table 1 above, we have 17 subjects for the 30min group (9+8),18 subjects for the 24h group (9 + 9) and 26 subjects for the 6h group (12 + 14). As Author response image 2 shows, the SCR patterns in the fear acquisition, extinction and test phases were similar to the results presented in the original figure.

Author response image 2.

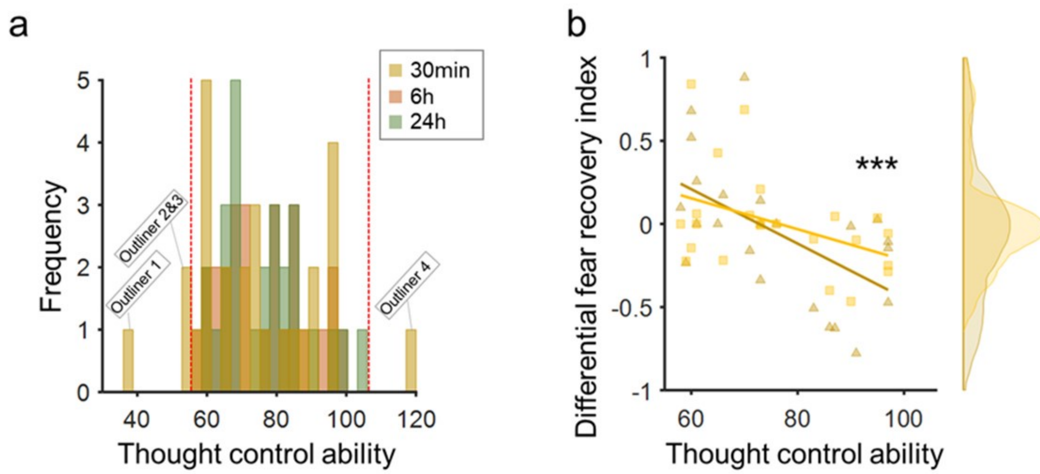


> *15G. Why is the range of scores in "thought control ability" different in the 30-minute group compared to the 6-hour and 24-hour groups? I am not just asking about the scale on the x-axis: I am asking why the actual distribution of the scores in thought control ability is wider for the 30-minute group?*

We went back and tested whether the TCAQ score variance was the same across three groups. We found that there was significant difference in the variance of the TCAQ score distribution across three groups ($F_{2.155}$ = 4.324, P = 0.015, Levene test). However, post-hoc analyses found that the variance of TCAQ is not significantly different between the 30min and 6h groups ($F_{26.25}$ = 0.4788, P = 0.0697), nor between the 30min and 24h groups (i>$F_{26.25}$ = 0.4692, P =

0.0625). To further validate our correlational results between the TCAQ score and the fear recovery index, we removed the TCAQ scores that were outside the TCAQ score range of the 6h & 24h groups from the 30min group (resulting in 4 “outliner” TCAQ scores in the 30min group, panel a in Author response image 3 below) and the Levene test confirmed that the variance of the TCAQ scores showed no difference across groups after removing the 4 “outliner” data points in the 30min group (i>$F_{2,147}$ = 0.74028, P = 0.4788). Even with the 4 “outliers” removed from the 30min group, the correlational analysis of the TCAQ scores and the fear recovery index still yielded significant result in the 30min group (*beta* = -0.0148, t = -3.731, P = 0.0006, see panel b below), indicating our results were not likely due to the inclusion of subjects with extreme TCAQ scores.

Author response image 3.



> *(16) During testing in each experiment, how were the various stimuli presented? That is, was the presentation order for the CS+ and CS- pseudorandom according to some constraint, as it had been in extinction? This information should be added to the method section.*

We mentioned the order of the stimuli in the testing phase in the methods section “… For studies 2 & 3, …a pseudo-random stimulus order was generated for fear acquisition and extinction phases of three groups with the rule that no same trial- type (CS1+, CS2+ and CS-) repeated more than twice. In the test phase, to exclude the possibility that the difference between CS1+ and CS2+ was simply caused by the presentation sequence of CS1+ and CS2+, half of the participants completed the test phase using a pseudo-random stimuli sequence and the identities of CS1+ and CS2+ reversed in the other half of the participants.”

> *(17) "These results are consistent with previous research which suggested that people with better capability to resist intrusive thoughts also performed better in motivated dementia in both declarative and associative memories."*
>
> ****Which parts of the present results are consistent with such prior results? It is not clear from the descriptions provided here why thought control ability should be related to the present findings or, indeed, past ones in other domains. This should be elaborated to make the connections clear.*

In the 30min group, we found that subjects’ TCAQ scores were negatively correlated with their fear recovery indices. That is, people with better capacity to resist intrusive thoughts were also less likely to experience the return of fear memory, which are consistent with previous results. Together with our brain stimulation results, the short-term amnesia is related to subject's cognitive control ability and intact dlPFC functions. It is because of these similarities that we propose that the short-term amnesia might be related to the automatic memory suppression mechanism originated from the declarative memory research. Since we have not provided all the evidence at this point of the results section, we briefly listed the connections with previous declarative and associative memory research.

> ***Reviewer #2 (Public Review):***
>
> *The fear acquisition data is converted to a differential fear SCR and this is what is analysed (early vs late). However, the figure shows the raw SCR values for CS+ and CS- and therefore it is unclear whether the acquisition was successful (despite there being an "early" vs "late" effect - no descriptives are provided).*

As the reviewer mentioned, the fear acquisition data was converted to a differential fear SCR and we conducted a two-way mixed ANOVA (reminder vs. no-reminder) x time (early vs. late part of fear acquisition) on the differential SCRs. We found a significant main effect of time (early vs. late; $F_{1,55}$ = 6.545, P = 0.013, η^2 = 0.106), suggesting successful fear acquisition in both groups. Fig. 1c also showed the mean differential SCR for the latter half of the acquisition phase in both the reminder and no-reminder groups and there was no significant difference in acquired SCRs between groups (early acquisition: t_{55} = -0.063, P = 0.950; late acquisition: t_{55} = -0.318, P = 0.751; Fig. 1c).

> *In Experiment 1 (Test results) it is unclear whether the main conclusion stems from a comparison of the test data relative to the last extinction trial ("we defined the fear recovery index as the SCR difference between the first test trial and the last extinction trial for a specific CS") or the difference relative to the CS- ("differential fear recovery index between CS+ and CS-"). It would help the reader assess the data if Figure 1e presents all the indexes (both CS+ and CS-). In addition, there is one sentence that I could not understand "there is no statistical difference between the differential fear recovery indexes between CS+ in the reminder and no reminder groups (P=0.048)". The p-value suggests that there is a difference, yet it is not clear what is being compared here. Critically, any index taken as a difference relative to the CS- can indicate recovery of fear to the CS+ or absence of discrimination relative to the CS-, so ideally the authors would want to directly compare responses to the CS+ in the reminder and no-reminder groups. The latter issue is particularly relevant in Experiment 2, in which the CS- seems to vary between groups during the test and this can obscure the interpretation of the result.*

In all the experiments, the fear recovery index (FRI) was defined as the SCR difference between the first test trial and the last extinction trial for any CS. Subsequently, the differential fear recovery index (FRI) was defined between the FRI of a specific CS+ and the FRI of the CS-. The differential FRI would effectively remove the non-specific time related effect (using the CS- FRI as the baseline). We have revised the text accordingly.

As we responded to reviewer #1, the CS- fear recovery indices (FIR) for the reminder and no-reminder groups were not statistically different (Kruskal-Wallis test $\chi^2_1 = 0.230, P = 0.631$, panel a, Author response image 1), though both groups showed significant fear recovery even in the CS- condition (Wilcoxon signed rank test, reminder: P = 0.0043, no-reminder: P = 0.0037, panel a). Next, we examined the mean SCR for CS- for the 30min, 6h and 24h groups in study 2 and found that there was indeed a group difference (one-way ANOVA, one-way ANOVA,$F_{2,76}$ = 5.3462, P = 0.0067, panel b), suggesting that the CS- SCR was influenced by the test time delay. We also tested the CS- SCR for the 4 groups in study 3 and found that across

TMS stimulation types (PFC vs. VER) and reminder types (reminder vs. no-reminder) the ANOVA analysis did not yield main effect of TMS stimulation type ($F_{1,71}$ = 0.322, P = 0.572) nor main effect of reminder type ($F_{1,71}$ = 0.0499, P = 0.824, panel c). We added the R-VER group results in study 3 (see panel c) to panel b and plotted the CS- SCR difference across 4 different test time points and found that CS- SCR decreased as the test-extinction delay increased (Jonckheere-Terpstra test, P = 0.00028). These results suggest a natural "forgetting" tendency for the CS- fear recovery index and highlight the importance of having the CS- as a control condition to compare the CS+ recovery index with (resulting in the Differential recovery index). Parametric and non-parametric analyses were adopted based on whether the data met the assumptions for the parametric analyses.

> *In Experiment 1, the findings suggest that there is a benefit of retrieval followed by extinction in a short-term reinstatement test. In Experiment 2, the same effect is observed on a cue that did not undergo retrieval before extinction (CS2+), a result that is interpreted as resulting from cue-independence, rather than a failure to replicate in a within-subjects design the observations of Experiment 1 (between-subjects). Although retrieval-induced forgetting is cue-independent (the effect on items that are suppressed [Rp-] can be observed with an independent probe), it is not clear that the current findings are similar. Here, both cues have been extinguished and therefore been equally exposed during the critical stage.*

We appreciate the reviewer's insight on this issue. Although in the discussion we raised the possibility of memory suppression to account for the short-term amnesia effect, we did not intend to compare our paradigm side-by-side with retrieval-induced forgetting. In our previous work (Wang et al., 2021), we reported that active suppression effect of CS+ related fear memory during the standard extinction training generalized to other CS+, yielding a cue-independent effect. In the current experiments, we did not implement active suppression; instead, we used the CS+ retrieval-extinction paradigm. It is thus possible that the CS+ retrieval cue may function to facilitate automatic suppression. Indeed, in the no-reminder group (standard extinction) of study 1, we did observe the return of fear expression, suggesting the critical role of CS+ reminder before the extinction training. Based on the results mentioned above, we believe our short-term amnesia results were consistent with the hypothesis that the retrieval CS+ (reminder) might prompt subjects to adopt an automatic suppress mechanism in the following extinction training, yielding cue-independent amnesia effects.

> *The findings in Experiment 2 suggest that the amnesia reported in Experiment 1 is transient, in that no effect is observed when the test is delayed by 6 hours. The phenomena whereby reactivated memories transition to extinguished memories as a function of the amount of exposure (or number of trials) is completely different from the phenomena observed here. In the former, the manipulation has to do with the number of trials (or the total amount of time) that the cues are exposed to. In the current study, the authors did not manipulate the number of trials but instead the retention interval between extinction and test. The finding reported here is closer to a "Kamin effect", that is the forgetting of learned information which is observed with intervals of intermediate length (Baum, 1968). Because the Kamin effect has been inferred to result from retrieval failure, it is unclear how this can be explained here. There needs to be much more clarity on the explanations to substantiate the conclusions.*

Indeed, in our studies, we did not manipulate the amount of exposure (or number of trials) but only the retention interval between extinction and test. Our results demonstrated that the retrieval-extinction protocol yielded the short-term amnesia on fear memory, qualitatively different from the reconsolidation related amnesia proposed in the previous literatures. After examining the temporal dynamics, cue-specificity and TCAQ association with the short-term amnesia, we speculated that the short-term effect might be related to an automatic suppression mechanism. Of course, further studies will be required to test such a hypothesis.

Our results might not be easily compared with the “Kamin effect”, a term coined to describe the “retention of a partially learned avoidance response over varying time intervals” using a learning-re-learning paradigm (Baum, 1968, Kamin, 1957). However, the retrieval-extinction procedure used in our studies was different from the learning-re-learning paradigm in the original paper (Kamin, 1957) and the reversal-learning paradigm the reviewer mentioned (Baum, 1968).

> *There are many results (Ryan et al., 2015) that challenge the framework that the authors base their predictions on (consolidation and reconsolidation theory), therefore these need to be acknowledged. Similarly, there are reports that failed to observe the retrieval-extinction phenomenon (Chalkia et al., 2020), and the work presented here is written as if the phenomenon under consideration is robust and replicable. This needs to be acknowledged.*

We thank the reviewer pointing out the related literature and have added a separate paragraph about other results in the discussion (as well as citing relevant references in the introduction) to provide a full picture of the reconsolidation theory to the audience:

“It should be noted that while our long-term amnesia results were consistent with the fear memory reconsolidation literatures, there were also studies that failed to observe fear prevention (Chalkia, Schroyens, et al., 2020; Chalkia, Van Oudenhove, et al., 2020; Schroyens et al., 2023). Although the memory reconsolidation framework provides a viable explanation for the long-term amnesia, more evidence is required to validate the presence of reconsolidation, especially at the neurobiological level (Elsey et al., 2018). While it is beyond the scope of the current study to discuss the discrepancies between these studies, one possibility to reconcile these results concerns the procedure for the retrieval-extinction training. It has been shown that the eligibility for old memory to be updated is contingent on whether the old memory and new observations can be inferred to have been generated by the same latent cause (Gershman et al., 2017; Gershman and Niv, 2012). For example, prevention of the return of fear memory can be achieved through gradual extinction paradigm, which is thought to reduce the size of prediction errors to inhibit the formation of new latent causes (Gershman, Jones, et al., 2013). Therefore, the effectiveness of the retrieval-extinction paradigm might depend on the reliability of such paradigm in inferring the same underlying latent cause. Furthermore, other studies highlighted the importance of memory storage per se and suggested that memory retention was encoded in the memory engram cell ensemble connectivity whereas the engram cell synaptic plasticity is crucial for memory retrieval (Ryan et al., 2015; Tonegawa, Liu, et al., 2015; Tonegawa, Pignatelli, et al., 2015). It remains to be tested how the cue-independent short-term and cue-dependent long-term amnesia effects we observed could correspond to the engram cell synaptic plasticity and functional connectivity among engram cell ensembles (Figure 6). This is particularly important, since the cue-independent characteristic of the short-term amnesia suggest that either different memory cues fail to evoke engram cell activities, or the retrieval-extinction training transiently inhibits connectivity among engram cell ensembles. Finally, SCR is only one aspect of the fear expression, how the retrieval-extinction paradigm might affect subjects’ other emotional (such as the startle response) and cognitive fear expressions such as reported fear expectancy needs to be tested in future studies since they do not always align with each other (Kindt et al., 2009; Sevenster et al., 2012, 2013).”

> *The parallels between the current findings and the memory suppression literature are speculated in the general discussion, and there is the conclusion that "the retrieval-extinction procedure might facilitate a spontaneous memory suppression process". Because one of the basic tenets of the memory suppression literature is that it reflects an*

> *"active suppression" process, there is no reason to believe that in the current paradigm, the same phenomenon is in place, but instead, it is "automatic". In other words, the conclusions make strong parallels with the memory suppression (and cognitive control) literature, yet the phenomena that they observed are thought to be passive (or spontaneous/automatic).*
>
> *Ultimately, it is unclear why 10 mins between the reminder and extinction learning will "automatically" suppress fear memories. Further down in the discussion, it is argued that "For example, in the well-known retrieval-induced forgetting (RIF) phenomenon, the recall of a stored memory can impair the retention of related long-term memory and this forgetting effect emerges as early as 20 minutes after the retrieval procedure, suggesting memory suppression or inhibition can occur in a more spontaneous and automatic manner". I did not follow with the time delay between manipulation and test (20 mins) would speak about whether the process is controlled or automatic.*

In our previous research, we showed that the memory suppression instruction together with the extinction procedure successfully prevented the return of fear expression in the reinstatement test trials 30mins after the extinction training (Wang et al., 2021). In the current experiments, we replaced the suppression instruction with the retrieval cue before the extinction training (retrieval-extinction protocol) and observed similar short-term amnesia effects. These results prompted us to hypothesize in the discussion that the retrieval cue might facilitate an automatic suppression process. We made the analogy to RIF phenomenon in the discussion to suggest that the suppression of (competing) memories could be unintentional and fast (20 mins), both of which were consistent with our results. We agree with the reviewer that this hypothesis is more of a speculation (hence in the discussion), and more studies are required to further test such a hypothesis. However, what we want to emphasize in this paper is the report of the short-term amnesia effects which were clearly not related to the memory reconsolidation effect in a variety of aspects.

> *Among the many conclusions, one is that the current study uncovers the "mechanism" underlying the short-term effects of retrieval extinction. There is little in the current report that uncovers the mechanism, even in the most psychological sense of the mechanism, so this needs to be clarified. The same applies to the use of "adaptive".*
>
> *Whilst I could access the data on the OFS site, I could not make sense of the Matlab files as there is no signposting indicating what data is being shown in the files. Thus, as it stands, there is no way of independently replicating the analyses reported.*

We have re-organized data on the OFS site, and they should be accessible now.

> *The supplemental material shows figures with all participants, but only some statistical analyses are provided, and sometimes these are different from those reported in the main manuscript. For example, the test data in Experiment 1 is analysed with a two-way ANOVA with the main effects of group (reminder vs no-reminder) and time (last trial of extinction vs first trial of the test) in the main report. The analyses with all participants in the sup mat used a mixed two-way ANOVA with a group (reminder vs no reminder) and CS (CS+ vs CS-). This makes it difficult to assess the robustness of the results when including all participants. In addition, in the supplementary materials, there are no figures and analyses for Experiment 3.*

We are sorry for the lack of clarity in the supplementary materials. We have supplementary figures Fig. S1 & S2 for the data re-analysis with all the responders (learners + non-learners). The statistical analyses performed on the responders in both figures yielded similar results as those in the main text. For other analyses reported in the supplementary materials, we specifically provided different analysis results to demonstrate the robustness of our results.

For example, to rule out the effects we observed in two-way ANOVA in the main text may be driven by the different SCR responses on the last extinction trial, we only tested the two-way ANOVA for the first trial SCR of test phase and these analyses provided similar results. Please note we did not include non-learners in these analyses (the texts of the supplementary materials).

Since we did not exclude any non-learners in study 3, all the results were already reported in the main text.

> *One of the overarching conclusions is that the "mechanisms" underlying reconsolidation (long term) and memory suppression (short term) phenomena are distinct, but memory suppression phenomena can also be observed after a 7-day retention interval (Storm et al., 2012), which then questions the conclusions achieved by the current study.*

As we stated before, the focus of the manuscript was to demonstrate a novel short-term fear amnesia effect following the retrieval-extinction procedure. We discussed memory suppression as one of the potential mechanisms for such a short-term effect. In fact, the durability of the memory suppression effect is still under debate. Although Storm et al. (2012) suggested that the retrieval-induced forgetting can persist for as long as a week, other studies, however, failed to observe long-term forgetting (after 24 hrs; (Carroll et al., 2007, Chan, 2009). It is also worth noting that Storm et al. (2012) tested RIF one week later using half of the items the other half of which were tested 5 minutes after the retrieval practice. Therefore, it can be argued that there is a possibility that the long-term RIF effect is contaminated by the test/re-test process on the same set of (albeit different) items at different time onsets (5mins & 1 week).

> ***Reviewer #3 (Public Review):***
>
> *(1) The entire study hinges on the idea that there is memory 'suppression' if (1) the CS+ was reminded before extinction and (2) the reinstatement and memory test takes place 30 minutes later (in Studies 1 & 2). However, the evidence supporting this suppression idea is not very strong. In brief, in Study 1, the effect seems to only just reach significance, with a medium effect size at best, and, moreover, it is unclear if this is the correct analysis (which is a bit doubtful, when looking at Figure 1D and E). In Study 2, there was no optimal control condition without reminder and with the same 30-min interval (which is problematic, because we can assume generalization between CS1+ and CS2+, as pointed out by the authors, and because generalization effects are known to be time-dependent). Study 3 is more convincing, but entails additional changes in comparison with Studies 1 and 2, i.e., applications of cTBS and an interval of 1 hour instead of 30 minutes (the reason for this change was not explained). So, although the findings of the 3 studies do not contradict each other and are coherent, they do not all provide strong evidence for the effect of interest on their own.*
>
> *Related to the comment above, I encourage the authors to double-check if this statement is correct: "Also, our results remain robust even with the "non-learners" included in the analysis (Fig. S1 in the Supplemental Material)". The critical analysis for Study 1 is a between-group comparison of the CS+ and CS- during the last extinction trial versus the first test trial. This result only just reached significance with the selected sample (p = .048), and Figures 1D and E even seem to suggest otherwise. I doubt that the analysis would reach significance when including the "non-learners" - assuming that this is what is shown in Supplemental Figure 1 (which shows the data from "all responded participants").*

Our subjects were categorized based on the criteria specified in supplementary table S1. More specifically, we excluded the non-responders (Mean CS SCR < 0.02 *uS* in the fear acquisition phase), and non-learners and focused our analyses on the learners. Non-responders were dismissed after day 1 (the day of fear acquisition), but both learners and non-learners finished the experiments. This fact gave us the opportunity to examine data for both the learners and the responders (learners + non-learners). What we showed in fig. 1D and E were differential SCRs (CS+ minus CS-) of the last extinction trials and the differential fear recovery indices (CS+ minus CS-), respectively. We have double checked the figures and both the learners (Fig. 1) and the responders (i.e. learners and non-learners, supplementary Fig. 1) results showed significant differences between the reminder and no-reminder groups on the differential fear recovery index.

> *Also related to the comment above, I think that the statement "suggesting a cue-independent short-term amnesia effect" in Study 2 is not correct and should read: "suggesting extinction of fear to the CS1+ and CS2+", given that the response to the CS+'s is similar to the response to the CS-, as was the case at the end of extinction. Also the next statement "This result indicates that the short-term amnesia effect observed in Study 2 is not reminder-cue specific and can generalize to the non-reminded cues" is not fully supported by the data, given the lack of an appropriate control group in this study (a group without reinstatement). The comparison with the effect found in Study 1 is difficult because the effect found there was relatively small (and may have to be double-checked, see remarks above), and it was obtained with a different procedure using a single CS+. The comparison with the 6-h and 24-h groups of Study 2 is not helpful as a control condition for this specific question (i.e., is there reinstatement of fear for any of the CS+'s) because of the large procedural difference with regard to the intervals between extinction and reinstatement (test).*

In Fig. 2e, we showed the differential fear recovery indices (FRI) for the CS+ in all three groups. Since the fear recovery index (FRI) was calculated as the SCR difference between the first test trial and the last extinction trial for any CS, the differential fear recovery indices (difference between CS+ FRI and CS- FRI) not significantly different from 0 should be interpreted as the lack of fear expression in the test phase. Since spontaneous recovery, reinstatement and renewal are considered canonical phenomena in demonstrating that extinction training does not really "erase" conditioned fear response, adding the no-reinstatement group as a control condition would effectively work as the spontaneous recovery group and the comparison between the reinstatement and no-instatement groups turns into testing the difference in fear recovery using different methods (reinstatement vs. spontaneous recovery).

> *(2) It is unclear which analysis is presented in Figure 3. According to the main text, it either shows the "differential fear recovery index between CS+ and CS-" or "the fear recovery index of both CS1+ and CS2+". The authors should clarify what they are analyzing and showing, and clarify to which analyses the ** and NS refer in the graphs. I would also prefer the X-axes and particularly the Y-axes of Fig. 3a-b-c to be the same. The image is a bit misleading now. The same remarks apply to Figure 5.*

We are sorry about the lack of clarity here. Figures 3 & 5 showed the correlational analyses between TCAQ and the differential fear recovery index (FRI) between CS+ and CS-. That is, the differential FRI of CS1+ (CS1+ FRI minus CS- FRI) and the differential FRI of CS2+ (CS2+ FRI minus CS- FRI).

We have rescaled both X and Y axes for figures 3 & 5 (please see the revised figures).

> *(3) In general, I think the paper would benefit from being more careful and nuanced in how the literature and findings are represented. First of all, the authors may be more careful when using the term 'reconsolidation'. In the current version, it is put forward as*

> *an established and clearly delineated concept, but that is not the case. It would be useful if the authors could change the text in order to make it clear that the reconsolidation framework is a theory, rather than something that is set in stone (see e.g., Elsey et al., 2018 (https://doi.org/10.1037/bul0000152), Schroyens et al., 2022 (https://doi.org/10.3758/s13423-022-02173-2)).*
>
> *In addition, the authors may want to reconsider if they want to cite Schiller et al., 2010 (https://doi.org/10.1038/nature08637), given that the main findings of this paper, nor the analyses could be replicated (see, Chalkia et al., 2020 (https://doi.org/10.1016/j.cortex.2020.04.017; https://doi.org/10.1016/j.cortex.2020.03.031).*

We thank the reviewer's comments and have incorporated the mentioned papers into our revised manuscript by pointing out the extant debate surrounding the reconsolidation theory in the introduction:

"Pharmacological blockade of protein synthesis and behavioral interventions can both eliminate the original fear memory expression in the long-term (24 hours later) memory test (Lee, 2008; Lee et al., 2017; Schiller et al., 2013; Schiller et al., 2010), resulting in the cue-specific fear memory deficit (Debiec et al., 2002; Lee, 2008; Nader, Schafe, & LeDoux, 2000). For example, during the reconsolidation window, retrieving a fear memory allows it to be updated through extinction training (i.e., the retrieval-extinction paradigm (Lee, 2008; Lee et al., 2017; Schiller et al., 2013; Schiller et al., 2010), but also see (Chalkia, Schroyens, et al., 2020; Chalkia, Van Oudenhove, et al., 2020; D. Schiller, LeDoux, & Phelps, 2020). "

As well as in the discussion:

"It should be noted that while our long-term amnesia results were consistent with the fear memory reconsolidation literatures, there were also studies that failed to observe fear prevention (Chalkia, Schroyens, et al., 2020; Chalkia, Van Oudenhove, et al., 2020; Schroyens et al., 2023). Although the memory reconsolidation framework provides a viable explanation for the long-term amnesia, more evidence is required to validate the presence of reconsolidation, especially at the neurobiological level (Elsey et al., 2018). While it is beyond the scope of the current study to discuss the discrepancies between these studies, one possibility to reconcile these results concerns the procedure for the retrieval-extinction training. It has been shown that the eligibility for old memory to be updated is contingent on whether the old memory and new observations can be inferred to have been generated by the same latent cause (Gershman et al., 2017; Gershman and Niv, 2012). For example, prevention of the return of fear memory can be achieved through gradual extinction paradigm, which is thought to reduce the size of prediction errors to inhibit the formation of new latent causes (Gershman, Jones, et al., 2013). Therefore, the effectiveness of the retrieval-extinction paradigm might depend on the reliability of such paradigm in inferring the same underlying latent cause. Furthermore, other studies highlighted the importance of memory storage per se and suggested that memory retention was encoded in the memory engram cell ensemble connectivity whereas the engram cell synaptic plasticity is crucial for memory retrieval (Ryan et al., 2015; Tonegawa, Liu, et al., 2015; Tonegawa, Pignatelli, et al., 2015). It remains to be tested how the cue-independent short-term and cue-dependent long-term amnesia effects we observed could correspond to the engram cell synaptic plasticity and functional connectivity among engram cell ensembles (Figure 6). This is particularly important, since the cue-independent characteristic of the short-term amnesia suggest that either different memory cues fail to evoke engram cell activities, or the retrieval-extinction training transiently inhibits connectivity among engram cell ensembles. Finally, SCR is only one aspect of the fear expression, how the retrieval-extinction paradigm might affect subjects' other emotional (such as the startle response) and cognitive fear expressions such as reported fear expectancy needs to be tested in future studies since they do not always align with each other (Kindt et al., 2009; Sevenster et al., 2012, 2013)."

> *Relatedly, it should be clarified that Figure 6 is largely speculative, rather than a proven model as it is currently presented. This is true for all panels, but particularly for panel c, given that the current study does not provide any evidence regarding the proposed reconsolidation mechanism.*

We agree with the reviewer that Figure 6 is largely speculative. We realize that there are still debates regarding the retrieval-extinction procedure and the fear reconsolidation hypothesis. We have provided a more elaborated discussion and pointed out that figure 6 is only a working hypothesis and more work should be done to test such a hypothesis:

"Although mixed results have been reported regarding the durability of suppression effects in the declarative memory studies (Meier et al., 2011; Storm et al., 2012), future research will be needed to investigate whether the short-term effect we observed is specifically related to associative memory or the spontaneous nature of suppression (Figure 6C)."

> *Lastly, throughout the paper, the authors equate skin conductance responses (SCR) with fear memory. It should at least be acknowledged that SCR is just one aspect of a fear response, and that it is unclear whether any of this would translate to verbal or behavioral effects. Such effects would be particularly important for any clinical application, which the authors put forward as the ultimate goal of the research.*

Again, we agree with the reviewer on this issue, and we have acknowledged that SCR is only one aspect of the fear response and caution should be exerted in clinical application:

"Finally, SCR is only one aspect of the fear expression, how the retrieval-extinction paradigm might affect subjects' other emotional (such as the startle response) and cognitive fear expressions such as reported fear expectancy needs to be tested in future studies since they do not always align with each other (Kindt et al., 2009; Sevenster et al., 2012, 2013)."

> *(4) The Discussion quite narrowly focuses on a specific 'mechanism' that the authors have in mind. Although it is good that the Discussion is to the point, it may be worthwhile to entertain other options or (partial) explanations for the findings. For example, have the authors considered that there may be an important role for attention? When testing very soon after the extinction procedure (and thus after the reminder), attentional processes may play an important role (more so than with longer intervals). The retrieval procedure could perhaps induce heightened attention to the reminded CS+ (which could be further enhanced by dlPFC stimulation)?*

We thank the reviewer for this suggestion and have added more discussion on the potential mechanisms involved. Unfortunately, since the literature on attention and fear recovery is rather scarce, it is even more of a speculation given our study design and results are mainly about subjects' skin conductance responses (SCR).

> *(5) There is room for improvement in terms of language, clarity of the writing, and (presentation of the) statistical analyses, for all of which I have provided detailed feedback in the 'Recommendations for the authors' section. Idem for the data availability; they are currently not publicly available, in contrast with what is stated in the paper. In addition, it would be helpful if the authors would provide additional explanation or justification for some of the methodological choices (e.g., the 18-s interval and why stimulate 8 minutes after the reminder cue, the choice of stimulation parameters), and comment on reasons for (and implications of) the large amount of excluded participants (>25%).*

We have addressed the data accessibility issue and added the justifications for the methodological choices as well as the excluded participants. As we mentioned in the manuscript and the supplementary materials, adding the non-learners into data analysis did not change the results. Since the non-responders discontinued after Day 1 due to their non-measurable spontaneous SCR signals towards different CS, it's hard to speculate whether or how the results might have changed. However, participants' exclusion rate in the SCR studies were relatively high (Hu et al., 2018, Liu et al., 2014, Raio et al., 2017, Schiller et al., 2010, Schiller et al., 2012, Wang et al., 2021). The non-responders were mostly associated with participants being tested in the winter in our tasks. Cold weather and dry skins in the winter are likely to have caused the SCR hard to measure (Bauer et al., 2022, Vila, 2004). Different intervals between the reinstating US (electric shock) and the test trials were used in the previous literature such as 10min (Schiller et al., 2010, Schiller et al., 2013) and 18 or 19s (Kindt and Soeter, 2018, Kindt et al., 2009, Wang et al., 2021). We stuck with the 18s reinstatement interval in the current experiment. For the cTBS stimulation, since the stimulation itself lasted less than 2mins, we started the cTBS 8min after the onset of reminder cue to ensure that any effect caused by the cTBS stimulation occurred during the hypothesized time window, where the old fear memory becomes labile after memory retrieval. All the stimulation parameters were determined based on previous literature, which showed that with the transcranial magnetic stimulation (TMS) on the human dorsolateral prefrontal cortex could disrupt fear memory reconsolidation (Borgomaneri et al., 2020, Su et al., 2022).

> *Finally, I think several statements made in the paper are overly strong in light of the existing literature (or the evidence obtained here) or imply causal relationships that were not directly tested.*

We have revised the texts accordingly.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *On numerous occasions there are typos and the autocorrect has changed "amnesia" for "dementia".*

We are sorry about this mistake and have revised the text accordingly.

> ***Reviewer #3 (Recommendations For The Authors):***
>
> **"Neither of the studies reported in this article was preregistered. The data for both studies are publicly accessible at https://osf.io/9agvk". This excerpt from the text suggests that there are 2 studies, but there are 3 in the paper. Also, the data are only accessible upon request, not publicly available. I haven't requested them, as this could de-anonymize me as a reviewer.*

We are sorry for the accessibility of the link. The data should be available to the public now.

> **Please refrain from causal interpretations when they are not supported by the data:*
>
> *- Figure 3 "thought-control ability only affected fear recovery"; a correlation does not provide causal evidence.*
>
> *- "establishing a causal link between the dlPFC activity and short-term fear amnesia." I feel this statement is too strong; to what extent do we know for sure what the applied stimulation of (or more correct: near) the dlPFC does exactly?*

We thank the reviewer for the suggestion and have changed the wording related to figure 3. On the other hand, we'd like to argue that the causal relationship between the dlPFC activity and short-term fear amnesia is supported by the results from study 3. Although the exact functional role of the TMS on dlPFC can be debated, the fact that the TMS stimulation on the dlPFC (compared to the vertex group) brought back the otherwise diminished fear memory expression can be viewed as the causal evidence between the dlPFC activity and short-term fear amnesia.

> **The text would benefit from language editing, as it contains spelling and grammar mistakes, as well as wording that is vague or inappropriate. I suggest the authors check the whole text, but below are already some excerpts that caught my eye:*
>
> *"preludes memory reconsolidation"; "old fear memory can be updated"; "would cause short-term memory deficit"; "the its functional coupling"; "Subjects (...) yielded more severe amnesia in the memory suppression tasks"; "memory retrieval might also precipitate a short-term amnesia effect"; "more SEVERE amnesia in the memory suppression tasks"; "the effect size of reinstatement effect"; "the previous literatures"; "towards different CS"; "failed to show SCR response to the any stimuli"; "significant effect of age of TMS"; "each subject' left hand"; "latter half trials"; "Differntial fear recovery"; "fear dementia"; "the fear reinstatement effects at different time scale is related to"; "fear reocery index"; "thought-control abiliites"; "performed better in motivated dementia"; "we tested that in addition to the memory retrieval cue (reminder), whether the"; "during reconsolidation window"; "consisitent with the short-term dementia"; "low level of shock (5v)"*

We thank the reviewer for thorough reading and sorry about typos in the manuscript. We have corrected typos and grammar mistakes as much as we can find.

> **In line with the remark above, there are several places where the text could still be improved.*
>
> *- The last sentence of the Abstract is rather vague and doesn't really add anything.*
>
> *- Please reword or clarify: "the exact functional role played by the memory retrieval remains unclear".*
>
> *- Please reword or clarify: "the unbinding of the old memory trace".*
>
> *- "suggesting that the fear memory might be amenable to a more immediate effect, in addition to what the memory reconsolidation theory prescribes" shouldn't this rather read "in contrast with"?*

We have modified the manuscript.

> *- In the Introduction, the authors state: "Specifically, memory reconsolidation effect will only be evident in the long-term (24h) memory test due to its requirement of new protein synthesis and is cue-dependent". They then continue about the more immediate memory update mechanisms that they want to study, but it is unclear from how the rationale is presented whether (and why (not)) they also expect this mechanism to be cue-dependent.*

Most of the previous studies on the fear memory reconsolidation using CS as the memory retrieval cues have demonstrated that the reconsolidation effect is cue-dependent (Kindt and Soeter, 2018, Kindt et al., 2009, Monfils et al., 2009, Nader et al., 2000, Schiller et al., 2013, Schiller et al., 2010, Xue et al., 2012). However, other studies using unconditioned stimulus retrieval-extinction paradigm showed that such protocol was able to prevent the return of fear memory expression associated with different CSs (Liu et al., 2014, Luo et al., 2015). In our task, we used CS+ as the memory retrieval cues and our results were consistent with results from previous studies using similar paradigms.

> *- "The effects of cTBS over the right dlPFC after the memory reactivation were assessed using the similar mixed-effect four-way ANOVA". Please clarify what was analyzed here.*
> *- "designing novel treatment of psychiatric disorders". Please make this more concrete or remove the statement.*

This sentence was right after a similar analysis performed in the previous paragraph. While the previous graph focused on how the SCRs in the acquisition phase were modulated by factors such as CS+ (CS1+ and CS2+), reminder (reminder vs. no-reminder), cTBS site (right dlPFC vs. vertex) and trial numbers, this analysis focused instead on the SCR responses in the extinction training phase. We have made the modifications as the reviewer suggested.

> **I have several concerns related to the (presentation) of the statistical analyses/results:*
> *- Some statistical analyses, as well as calculation of certain arbitrary indices (e.g., differential fear recovery index) are not mentioned nor explained in the Methods section, but only mentioned in the Results section.*

We have added the explanation of the differential fear recovery index into the methods section:

"To measure the extent to which fear returns after the presentation of unconditioned stimuli (US, electric shock) in the test phase, we defined the fear recovery index as the SCR difference between the first test trial and the last extinction trial for a specific CS for each subject. Similarly, in studies 2 and 3, differential fear recovery index was defined as the difference between fear recovery indices of CS+ and CS- for both CS1+ and CS2+."

> *- Figure 1C-E: It is unclear what the triple *** mean. Do they have the same meaning in Figure 1C and Figure 1E? I am not sure that that makes sense. The meaning is not explained in the figure caption (I think it is different from the single asterisk*) and is not crystal clear from the main text either.*

We explained the triple *** in the figure legend (Fig. 1): ***P < 0.001. The asterisk placed within each bar in Figure 1C-E indicates the statistical results of the post-hoc test of whether each bar was significant. For example, the *** placed inside bars in Figure 1E indicates that the differential fear recovery index is statistically significant in the no-reminder group (P < 0.001).

> *- Supplemental Figure 1: "with all responded participants" Please clarify how you define 'responded participants' and include the n's.*

We presented the criteria for both the responder/non-responder and the learner/non-learner in the table of the supplementary materials and reported the number of subjects in each category (please see supplement Table 1).

> *- "the differential SCRs (difference between CS+ and CS-) for the CS+". Please clarify what this means and/or how it is calculated exactly.*

Sorry, it means the difference between the SCRs invoked by CS+ and CS- for both CS1+ (CS1+ minus CS-) and CS2+ (CS2+ minus CS-).

> **I suggest that the authors provide a bit more explanation about the thought-control ability questionnaire. For example, the type of items, etc, as this is not a very commonly used questionnaire in the fear conditioning field.*

We provided a brief introduction to the thought-control ability questionnaire in the methods section:

"The control ability over intrusive thought was measured by the 25-item Thought-Control Ability Questionnaire (TCAQ) scle(30). Participants were asked to rate on a five-point Likert-type scale the extent to which they agreed with the statement from 1 (*completely disagree*) to 5 (*completely agree*). At the end of the experiments, all participants completed the TCAQ scale to assess their perceived control abilities over intrusive thoughts in daily life(17)."

We have added further description of the item types to the TCAQ scale.

> **The authors excluded more than 25% of the participants. It would be interesting to hear reasons for this relatively large number and some reflection on whether they think this selection affects their results (e.g., could being a (non)responder in skin conductance influence the susceptibility to reactivation-extinction in some way?).*

Participants exclusion rate in the SCR studies were relatively high (Hu et al., 2018, Liu et al., 2014, Raio et al., 2017, Schiller et al., 2010, Schiller et al., 2012, Wang et al., 2021). The non-responders were mostly associated with participants being tested in the winter in our tasks. Cold weather and dry skins in the winter are likely to have caused the SCR hard to measure (Bauer et al., 2022, Vila, 2004).

> **Minor comments that the authors may want to consider:*
>
> *- Please explain abbreviations upon first use, e.g., TMS.*
>
> *- In Figure 6, it is a bit counterintuitive that the right Y-axis goes from high to low.*

We added the explanation of TMS:

"Continuous theta burst stimulation (cTBS), a specific form of repetitive transcranial magnetic stimulation (rTMS)…"

We are sorry and agree that the right Y-axis was rather counterintuitive. However, since the direction of the fear recovery index (which was what we measured in the experiment) and the short/long-term amnesia effect are of the opposite directions, plotting one index from low to high would inevitably cause the other index to go from high to low.

Reference:

Anderson, M. C. and Floresco, S. B. 2022. Prefrontal-hippocampal interactions supporting the extinction of emotional memories: The retrieval stopping model. Neuropsychopharmacology, 47, 180-195.

Anderson, M. C. and Green, C. 2001. Suppressing unwanted memories by executive control. Nature, 410, 366-9.

Bauer, E. A., Wilson, K. A. and Macnamara, A. 2022. 3.03 - cognitive and affective psychophysiology. In: ASMUNDSON, G. J. G. (ed.) Comprehensive clinical psychology (second edition). Oxford: Elsevier.

Baum, M. 1968. Reversal learning of an avoidance response and the kamin effect. J Comp Physiol Psychol, 66, 495-7.

Borgomaneri, S., Battaglia, S., Garofalo, S., Tortora, F., Avenanti, A. and Di Pellegrino, G. 2020. State-dependent tms over prefrontal cortex disrupts fear-memory reconsolidation and prevents the return of fear. Curr Biol, 30, 3672-3679.e4.

Cain, C. K., Blouin, A. M. and Barad, M. 2003. Temporally massed cs presentations generate more fear extinction than spaced presentations. J Exp Psychol Anim Behav Process, 29, 323-33.

Carroll, M., Campbell-Ratcliffe, J., Murnane, H. and Perfect, T. 2007. Retrieval-induced forgetting in educational contexts: Monitoring, expertise, text integration, and test format. European Journal of Cognitive Psychology, 19, 580-606.

Chan, J. C. K. 2009. When does retrieval induce forgetting and when does it induce facilitation? Implications for retrieval inhibition, testing effect, and text processing. Journal of Memory and Language, 61, 153-170.

Gagnepain, P., Henson, R. N. and Anderson, M. C. 2014. Suppressing unwanted memories reduces their unconscious influence via targeted cortical inhibition. Proc Natl Acad Sci U S A, 111, E1310-9.

Gershman, S. J., Jones, C. E., Norman, K. A., Monfils, M. H. and Niv, Y. 2013. Gradual extinction prevents the return of fear: Implications for the discovery of state. Front Behav Neurosci, 7, 164.

Gershman, S. J., Monfils, M. H., Norman, K. A. and Niv, Y. 2017. The computational nature of memory modification. Elife, 6.

Hu, J., Wang, W., Homan, P., Wang, P., Zheng, X. and Schiller, D. 2018. Reminder duration determines threat memory modification in humans. Sci Rep, 8, 8848.

Kamin, L. J. 1957. The retention of an incompletely learned avoidance response. J Comp Physiol Psychol, 50, 457-60.

Kindt, M. and Soeter, M. 2018. Pharmacologically induced amnesia for learned fear is time and sleep dependent. Nat Commun, 9, 1316.

Kindt, M., Soeter, M. and Vervliet, B. 2009. Beyond extinction: Erasing human fear responses and preventing the return of fear. Nat Neurosci, 12, 256-8.

Liu, J., Zhao, L., Xue, Y., Shi, J., Suo, L., Luo, Y., Chai, B., Yang, C., Fang, Q., Zhang, Y., Bao, Y., Pickens, C. L. and Lu, L. 2014. An unconditioned stimulus retrieval extinction procedure to prevent the return of fear memory. Biol Psychiatry, 76, 895-901.

Luo, Y.-X., Xue, Y.-X., Liu, J.-F., Shi, H.-S., Jian, M., Han, Y., Zhu, W.-L., Bao, Y.-P., Wu, P., Ding, Z.-B., Shen, H.-W., Shi, J., Shaham, Y. and Lu, L. 2015. A novel ucs memory retrieval-extinction procedure to inhibit relapse to drug seeking. Nature Communications, 6, 7675.

Monfils, M. H., Cowansage, K. K., Klann, E. and Ledoux, J. E. 2009. Extinction-reconsolidation boundaries: Key to persistent attenuation of fear memories. Science, 324, 951-5.

Nader, K., Schafe, G. E. and Le Doux, J. E. 2000. Fear memories require protein synthesis in the amygdala for reconsolidation after retrieval. Nature, 406, 722-6.

Raio, C. M., Hartley, C. A., Orederu, T. A., Li, J. and Phelps, E. A. 2017. Stress attenuates the flexible updating of aversive value. Proc Natl Acad Sci U S A, 114, 11241-11246.

Schiller, D., Kanen, J. W., Ledoux, J. E., Monfils, M. H. and Phelps, E. A. 2013. Extinction during reconsolidation of threat memory diminishes prefrontal cortex involvement. Proc Natl Acad Sci U S A, 110, 20040-5.

Schiller, D., Monfils, M. H., Raio, C. M., Johnson, D. C., Ledoux, J. E. and Phelps, E. A. 2010. Preventing the return of fear in humans using reconsolidation update mechanisms. Nature, 463, 49-53.

Schiller, D., Raio, C. M. and Phelps, E. A. 2012. Extinction training during the reconsolidation window prevents recovery of fear. J Vis Exp, e3893.

Su, S., Deng, J., Yuan, K., Gong, Y., Zhang, Y., Li, H., Cao, K., Huang, X., Lin, X., Wu, P., Xue, Y., Bao, Y., Shi, J., Shi, L. and Lu, L. 2022. Continuous theta-burst stimulation over the right dorsolateral prefrontal cortex disrupts fear memory reconsolidation in humans. iScience, 25, 103614.

Vila, J. 2004. Psychophysiological assessment. In: SPIELBERGER, C. D. (ed.) Encyclopedia of applied psychology. New York: Elsevier.

Wang, Y., Zhu, Z., Hu, J., Schiller, D. and Li, J. 2021. Active suppression prevents the return of threat memory in humans. Commun Biol, 4, 609.

Xue, Y. X., Luo, Y. X., Wu, P., Shi, H. S., Xue, L. F., Chen, C., Zhu, W. L., Ding, Z. B., Bao, Y. P., Shi, J., Epstein, D. H., Shaham, Y. and Lu, L. 2012. A memory retrieval-extinction procedure to prevent drug craving and relapse. Science, 336, 241-5.

Zhu, Z., Anderson, M. C. and Wang, Y. 2022. Inducing forgetting of unwanted memories through subliminal reactivation. Nature communications, 13, 6496-6496.

https://doi.org/10.7554/eLife.98652.2.sa0